

Soft Skills, Hard Times: Why New College Graduates Are Falling Behind

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Abstract

The labor market outcomes for young college graduates have been worsening since the Great Recession. We document that this secular trend coincides with a rise in demand for soft skills by employers. We propose that this shift in skill demand disproportionately affects young workers, who have had less opportunity to accumulate these skills through on-the-job experience. To quantify this mechanism, we build a general equilibrium search model with heterogeneous agents and on-the-job skill accumulation, calibrated to the US labor market. A rise in demand for soft skills explains about one-third of the increase in the unemployment rate of young college graduates in this model.

Keywords: soft skills, youth unemployment, experience premium

JEL classification: J23, J24, J64

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The purpose of college education is to prepare the youth for the labor force by providing them with the skills necessary to achieve a successful career. However, young college graduates in recent times are finding it increasingly difficult to get jobs relative to the average US worker. In this paper, we discuss why a college education in its current form is no longer enough to ensure success in the labor market.

Deming (2017), Lindqvist and Vestman (2011) and Edin et al. (2022) have documented the growing importance of soft skills in the labor market. We claim that these skills are primarily accumulated on the job. This implies that even if young college graduates have adequate technical skills, they may lack the soft skills needed by employers. As the demand for soft skills by employers increases, young college graduates are left at a disadvantage, experiencing worse labor market outcomes as a result.

Using the Current Population Survey (CPS) and data on the near-universe of online job postings in the US obtained from Lightcast, we document three empirical facts that motivate our interpretation. First, we show that historically, young college graduates were less likely to be unemployed than the average US worker.¹ In other words, a college education was associated with better labor market outcomes for workers in the past. However, we find that since 2011, the unemployment rate of young college graduates has been worsening, and since 2016, they are more likely to be unemployed than the average US worker.

Next, we show that coinciding with the rising unemployment rate of young college graduates, there has been an increase in the demand for soft skills by employers in the US labor market. For this exercise, we identify soft skills and technical skills in our job postings dataset using a combination of large language models and manual review, and we define skill demand as the average number of soft (technical) skills required in each job posting. Using this skill classification, we find that between 2011 and 2024, technical skill demand increased by 6%, possibly reflecting an environment where work is getting more specialized. However, during the same time period, soft skill demand increased by 16%.

Finally, we provide evidence that young workers fare worse in occupations that require more soft skills. Specifically, we show that the wage gap between young and older workers widens by about 3 percent for each additional soft skill required in a job. This holds for both college educated workers and non-college educated workers. Interestingly, the relationship between technical skill demand and the wage gap depends on the education level of workers. For the non college educated sample, the wage gap between young and older workers also widens as technical skill demand increases. However, in the college educated sample, this wage gap becomes narrower in jobs that require more technical skills.

The wage gap patterns above suggest that although young college graduates have the relevant technical skills, they lack the soft skills required by employers. This can happen if soft skills are primarily accumulated on the job, which would imply that younger workers have had less opportunity to acquire them than older workers. Together, our three empirical facts indicate that an increase in the demand for soft skills can be an important factor behind the rising unemployment rate of young college graduates.

Using a general equilibrium search-and-matching model, we quantify how much of the rise

¹We define young workers as those aged 25 or younger. Older workers are between the ages of 26 and 55 years.

in the unemployment rate of young college graduates can be explained by the rise in demand for soft skills. Our model contains heterogeneous workers and firms, with on-the-job search and aggregate productivity shocks. Firms post vacancies, and both employed and unemployed workers receive job offers at random. Match creation is efficient, and wage determination happens through the sequential auctions framework developed by [Postel-Vinay and Robin \(2002\)](#). Workers can accumulate skills while employed. However, unemployment leads to skill depreciation.

Despite having heterogeneous workers and firms along with aggregate shocks, the model remains tractable. Worker flows and wage determination only depend on the match surplus, and the match surplus is independent of the current distributions of workers, vacancies, and matches ([Lise and Robin, 2017](#)). This means we can first solve for the surplus function, and then obtain the above distributions given the initial conditions. We calibrate this model to match key aggregate moments from the US labor market between 1951 and 2012.

In our counterfactual exercise, we incorporate skill biased technical change to generate an increase in skill demand by employers, equivalent to the rise in soft skill demand that we observe in the data over the 2011-2024 period. Due to skill biased technical change, all workers become productive, but high-skilled workers experience a larger increase in their productivity. This leads to a shift in the vacancy distribution towards higher skilled workers, which is the model equivalent of the rise in skill demand. Consequently, low skilled young workers face a lower share of viable vacancies, and their unemployment rate rises as a result. We show that this calibrated increase in skill demand generates approximately 36% of the observed increase in the unemployment rate of young college graduates.

Our paper contributes to the literature documenting the declining returns to cognitive skills and the rising importance of noncognitive skills. [Castex and Kogan Dechter \(2014\)](#) document that cognitive skills had a 30-50% larger effect on wages in the 1980s compared to the 2000s. [Beaudry et al. \(2016\)](#) and [Brynjolfsson and McAfee \(2014\)](#) attribute this to the rise in the adoption of IT, as computers can substitute for human workers by easily doing tasks requiring cognitive skills.

During the same time period, some papers show that noncognitive (or soft) skills have become an important determinant of labor market outcomes. Using Swedish military enlistment data, [Lindqvist and Vestman \(2011\)](#) observe that low levels of noncognitive skills are a strong predictor of failure in the labor market (i.e. unemployment), while high levels of cognitive (or technical) skills are necessary for success (i.e. high wages). Moreover, [Deming \(2017\)](#) in the case of the US, and [Edin et al. \(2022\)](#) in the context of Sweden show that the returns to noncognitive skills have increased over time. Our paper contributes to this literature by highlighting that noncognitive skills are typically accumulated on the job, making young workers the ones who suffer in the labor market.

Another set of papers discusses wage stagnation and the reduced demand for junior talent. [Guvenen et al. \(2022\)](#) document that median lifetime earnings in the US have stagnated. They show that this occurs primarily due to the stagnation of wages at the time of labor market entry. The [New York Federal Reserve](#) keeps track of the labor market outcomes of young workers, and it shows that wages have remained stagnant in recent times as well. [Bianchi and Paradisi \(2024\)](#) and [Citino et al. \(2026\)](#) document the difficulties faced by junior talent, which

has intensified in recent times due to remote work (Schubert, 2026; Emanuel et al., 2026).

Brynjolfsson et al. (2025) and Hosseini Maasoum and Lichtinger (2025) attribute the worsening outcomes of young college graduates to AI, while Emanuel et al. (2026) propose remote work as an explanation. AI can shift employer demand towards older workers, since AI may not have the context specific knowledge that comes with experience, although it can easily do intensive cognitive tasks that were previously assigned to young workers. Remote work can harm the productivity of young workers, as Emanuel et al. (2026) show that proximity to experienced workers is important for young, inexperienced workers to learn how their firm operates. Even though both AI and remote work are important determinants of young workers' outcomes in recent years, these reasons cannot explain why this worsening trend exists even in pre-COVID times. Our paper provides another mechanism at play: the rising demand for soft skills that are primarily accumulated on the job. This skill demand mechanism should be considered in conjunction with the AI and remote work mechanisms to explain the current difficulties faced by young workers.

The closest paper to ours is Wright (2026). They document a 50% decline in the employment rate of teenagers since the 1990s, and they attribute it to crowding out by adult workers whose own labor market opportunities have worsened. Although similar in nature, our paper focuses on young college graduates, and we attribute their worsening labor market outcomes to rising demand for soft skills.

The rest of the paper is organized as follows. First, we describe our data sources and the classification of soft and technical skills in Section I. We then document our empirical results in Section II. In Section III, we describe our model, and discuss its parametrization in Section IV. Section V contains our counterfactual exercise. We conclude in Section VI.

I Data

To understand the labor market outcomes of young college graduates, we need to measure both supply and demand side outcomes in the labor market. We use data from the Current Population Survey (CPS) to measure worker outcomes on the supply side, and we use online job postings data from Lightcast to measure employer side demand. We also use occupational data from the Occupational Information Network (O*NET) to assess the skill measures. Appendix A1 describes the data construction and reports additional descriptive tables and figures.

I.A Workers and Labor-Market Outcomes

We use the CPS to compare the employment and earnings of young college graduates with those of older graduates and workers without a college degree. College graduates are workers with a bachelor's degree or higher. We use the CPS monthly survey for employment outcomes and the Monthly Outgoing Rotation Group (MORG) for earnings data. The time period of focus is 2011–2024, the common period covered by the CPS and Lightcast. In Appendix A1.1 we describe the CPS data and sample construction, and report descriptive statistics for the sample.

I.B Job Postings and Skill Requirements

Lightcast, formerly Burning Glass Technologies, collects online job postings and extracts the skills employers request, along with information on the job and employer. These postings allow us to measure advertised requirements within occupations and follow their changes over time.

We distinguish soft skills from technical skills using O*NET’s definition of soft skills: interpersonal and thinking skills used to work with others and perform effectively in the workplace (O*NET, 2026). Our soft-skill category includes communication, management, negotiation, planning, and problem solving, as well as personal attributes such as self-motivation and attention to detail. The classification describes what employers request; it does not distinguish where or when workers acquire these capabilities. Skills outside this category are classified as technical. Appendix A1.2 describes the classification; Table 1 lists common skills in each category. We also use the classification in Deming (2017) and find that the results are robust to this alternative.

To capture skill demand, we propose the following measure: average number of distinct skills required per posting in an occupation and month, calculated separately for soft and technical skills. This measure captures the breadth of advertised requirements and gives equal weight to each skill.²

Figure 1 shows how the two measures vary across occupations over 2011–2024. Wholesale and manufacturing sales representatives selling nontechnical products and lawyers list similar numbers of technical skills per posting, but sales representatives list more soft skills. Sales managers also list more soft skills than accountants and auditors. Heavy truck drivers and computer network architects illustrate the low and high ends of the technical-skill axis.

We complement skill counts with a large language model (LLM) classification of individual job advertisements. Reading the full text allows the model to consider how a skill enters the job’s duties. The model assigns separate high or low ratings to soft and technical requirements. A high rating indicates that the skill type is important for the main duties, repeatedly emphasized, or explicitly required at an advanced level; generic mentions alone receive a low rating. An advertisement can receive a high rating on both dimensions. This measure captures the substantive role of skills in the advertised job, rather than treating every mention equally. O*NET ratings help interpret the duties, but the classification must be supported by the advertisement itself. Appendix A2 describes the text sample, scoring procedure, and construction of the intensity series.³

²O*NET rates the importance and level of skills required in each occupation; Appendix A3 compares these ratings with our skill counts. We also construct a separate measure of skill intensity by asking an LLM to read each advertisement and rate its soft and technical requirements as high or low. O*NET ratings provide context, while the advertisement’s duties and qualifications provide the evidence for the rating. Appendix A2 describes this procedure.

³We compare advertised skill counts with O*NET ratings of the importance and level of skills required in an occupation. Occupations with higher O*NET ratings tend to have higher advertised skill counts. Appendix A3 reports the comparison; Appendix A1.2 examines posting coverage, occupational composition, and the distribution of skill requirements.

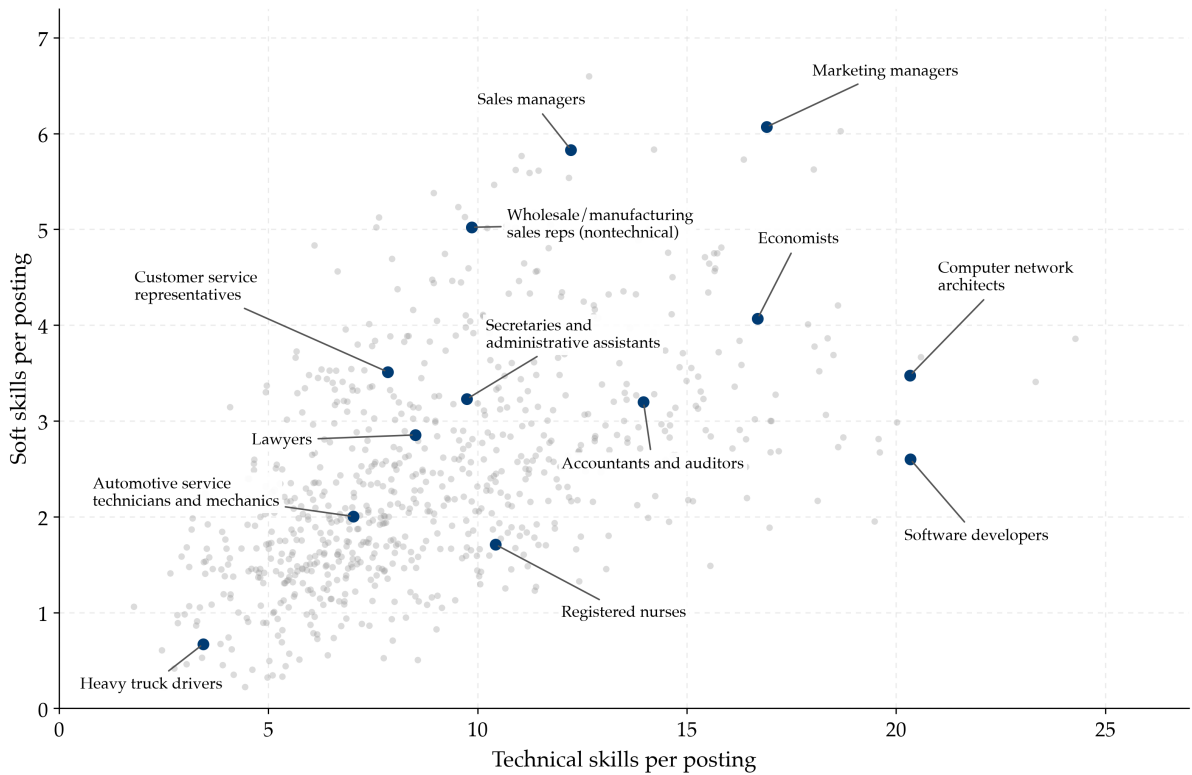
Table 1: Most Common Soft and Technical Skills in Lightcast Job Postings, 2011–2024

Panel A: Soft Skills					
#	Skill	Share (%)	#	Skill	Share (%)
1	Communication	33.76	16	Time Management	5.72
2	Customer Service	28.97	17	Presentations	5.64
3	Management	25.59	18	Troubleshooting (Problem Solving)	5.55
4	Sales	19.70	19	Professionalism	5.29
5	Operations	16.93	20	Selling Techniques	5.12
6	Leadership	15.63	21	Teamwork	4.99
7	Problem Solving	11.51	22	Prioritization	4.46
8	Detail Oriented	11.48	23	Coaching	4.44
9	Planning	10.16	24	Decision Making	3.98
10	Interpersonal Communications	8.18	25	Cleanliness	3.95
11	Coordinating	6.96	26	Innovation	3.87
12	Multitasking	6.83	27	Accountability	3.68
13	Verbal Communication Skills	6.78	28	Influencing Skills	3.23
14	Organizational Skills	6.71	29	Analytical Skills	3.12
15	Self-Motivation	6.33	30	Mentorship	2.99

Panel B: Technical Skills					
#	Skill	Share (%)	#	Skill	Share (%)
1	Writing	10.87	16	Microsoft PowerPoint	4.32
2	Microsoft Excel	8.48	17	Mathematics	4.25
3	Valid Driver’s License	8.33	18	Warehousing	4.05
4	Registered Nurse (RN)	8.15	19	Basic Life Support (BLS) Certification	4.04
5	Project Management	7.66	20	Microsoft Outlook	4.03
6	Merchandising	7.43	21	Computer Science	4.02
7	Microsoft Office	7.41	22	Restaurant Operation	3.77
8	Research	7.01	23	Workflow Management	3.04
9	Marketing	6.76	24	SQL (Programming Language)	2.98
10	Computer Literacy	6.62	25	Microsoft Word	2.89
11	Nursing	6.61	26	Product Knowledge	2.82
12	Accounting	6.08	27	Process Improvement	2.80
13	Auditing	5.85	28	Billing	2.80
14	Scheduling	5.77	29	Invoicing	2.74
15	Finance	4.98	30	Cash Register	2.71

Notes: This table lists the 30 most common skills in each category in Lightcast postings from 2011–2024. The categories come from an earlier classification of individual skill labels using LLMs and manual review. They differ from the taxonomy rules used in the skill-count figures, described in Appendix A1.2. Each share is the percentage of postings requesting that skill. The separate LLM intensity measure assigns high or low ratings to the requirements described in a full advertisement.

Figure 1: Soft and Technical Requirements across Occupations



Notes: Each point represents one of the six-digit SOC occupations in Lightcast. The horizontal and vertical axes show the average numbers of technical and soft skills per posting, respectively, pooling 307,456,240 postings from 2011–2024. Skill mentions and postings are summed within occupation before calculating ratios, giving each posting equal weight. Soft skills follow the taxonomy rules in Appendix A1.2; technical skills comprise all remaining labels, including certifications. Blue points mark thirteen illustrative occupations.

II Empirical Analysis

In this section, we examine whether the rising demand for soft skills can help explain the worsening labor market outcomes of young college graduates. We first document the rise in their unemployment rate relative to the overall rate and the increase in advertised soft-skill requirements since 2011. We then ask whether the wage gap between young and older workers is larger in occupations that require more soft skills, and how this relationship differs for technical skills.

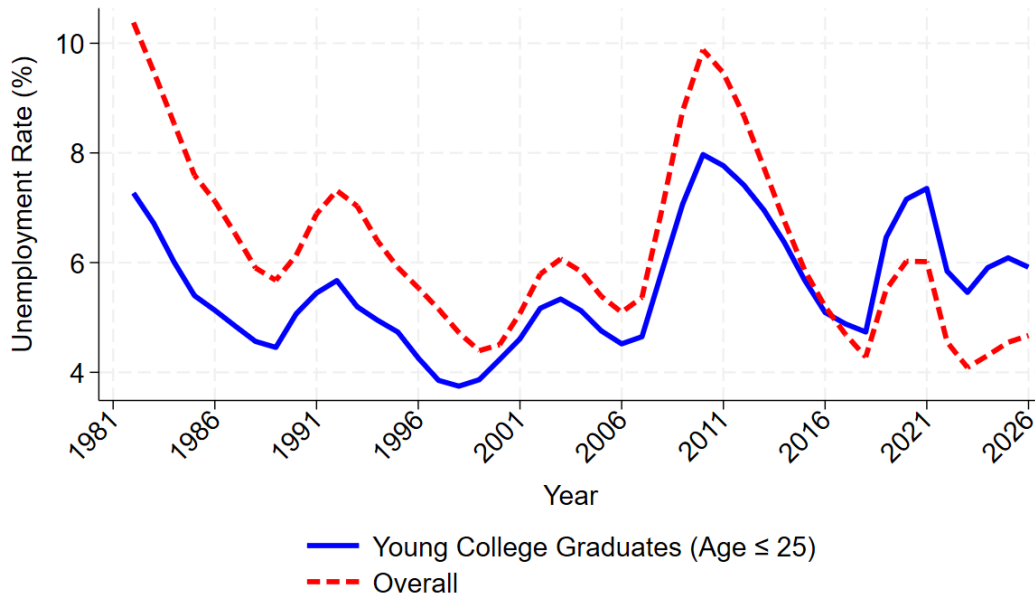
Work experience may be particularly valuable for requirements that involve judgment about people and situations. Negotiating with clients and managing work require anticipating others' responses and deciding how to proceed when plans fail. Experience can provide feedback on which actions work under different circumstances. In the framework of [Deming \(2021\)](#), workers use past experience to improve their predictions about the consequences of decisions. Our hypothesis is that rising demand for requirements that benefit from such experience can disadvantage recent entrants. This mechanism concerns part of our broad soft-skill category, which also contains attributes and skills developed before employment.

Young college graduates have been experiencing a decline in their employment advantage relative to the broader labor force. [Figure 2](#) compares the unemployment rate of graduates aged 25 or younger with the overall rate in our CPS sample. Between 1981 and 2011, young college graduates consistently experienced a relatively low unemployment rate, indicating that they were attractive to employers and had an easier time finding jobs compared to the rest of the population. However, since 2011, the gap between these 2 lines has been shrinking, and since 2015, the youth unemployment rate has been higher than the overall unemployment rate.⁴ Furthermore, labor market transitions show that young graduates have lost their initial job-finding advantage ([Appendix A5](#)). In 2011–2015, unemployed graduates aged 21–29 were 6 percentage points more likely to find employment the following month than graduates aged 30–55. This advantage narrowed by about 65% to 2 points in 2016–2019 and disappeared by 2022–2024, as older graduates' job-finding rate rose while young graduates' rate changed little. The gap in transitions from employment to unemployment remained nearly unchanged, and direct entry from outside the labor force into employment increased for young graduates.⁵ These reversals suggest that changes in the underlying economy have disproportionately affected young college graduates. Given that this trend starts from 2011, it cannot be explained by the AI boom or remote work alone, which have only been prevalent in the post-COVID period.

⁴Over 2011–2024, young graduates' unemployment advantage eroded even as the share with STEM bachelor's fields increased ([Appendix Figure A9](#)). Unemployment among young graduates in most of the fields shown, including STEM, increased relative to the overall rate ([Appendix Figure A10](#)). This weighs against a simple explanation in which young graduates are struggling because they increasingly choose fields outside STEM, and motivates examining whether employers increasingly require workplace skills that a college degree alone may not provide. These ACS series describe completed bachelor's fields among degree holders aged 18–25, not enrollment; field shares do not directly measure workers' skills or establish the effect of soft-skill demand.

⁵[Cline and Kaymak \(2025\)](#) document a similar erosion of the job-finding advantage when comparing young college and high-school graduates. [Guvenen et al. \(2022\)](#) show that lower earnings at labor market entry account for much of the decline in median lifetime earnings across male cohorts.

Figure 2: The Unemployment Rate Of Young College Graduates Is Rising



Notes: This figure shows the time series for the unemployment rate of young college graduates, and for the overall unemployment rate in the US (Current Population Survey, 1982-2025). A worker is considered young if she is aged 25 or lower. College graduates are defined as those workers who report holding a bachelor's degree or higher.

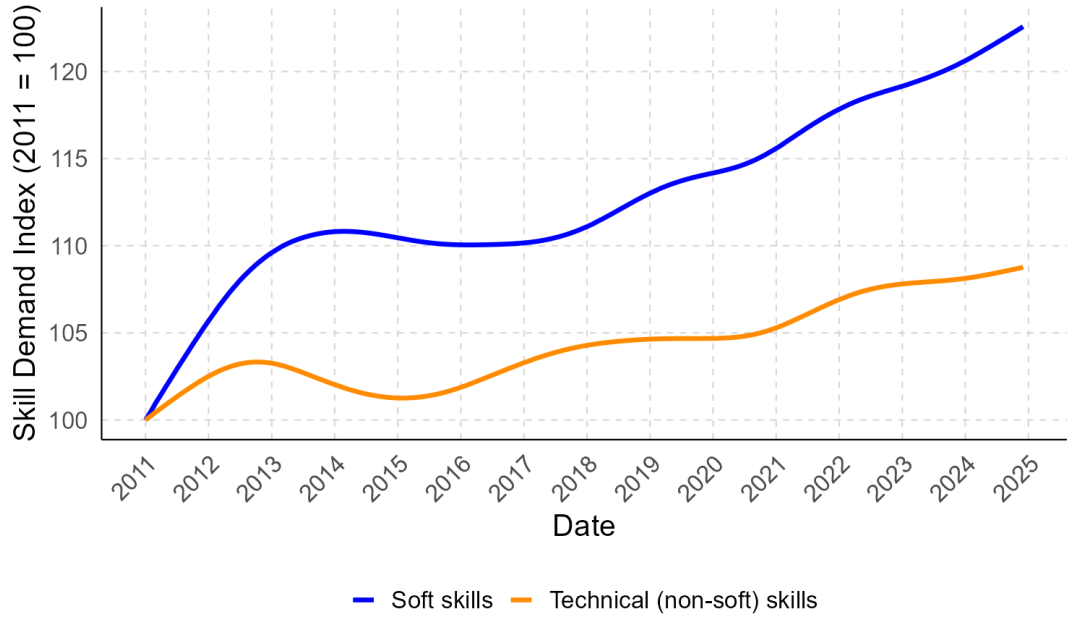
Young workers without a college degree, however, have fared differently: their unemployment rate has fallen relative to the overall rate (Appendix Figure A13). One possible explanation is that they have accumulated more work experience by age 25 than college graduates. If employers increasingly value skills acquired on the job, this difference in experience could help explain why the two groups' labor market outcomes have diverged.

To assess whether changing skill requirements could contribute to this divergence, we turn to Lightcast job postings. We first examine whether employers have increased their demand for soft skills relative to technical skills. Figure 3 plots the average numbers of soft and technical skills per posting. Both series rise over 2011–2024, with a somewhat larger proportional increase in soft skills. This reflects both changes in requirements within occupations and changes in the occupations represented in job postings.

To distinguish changes within occupations from shifts in the composition of postings, Table 2 reports regressions of log skill counts per posting on a linear time trend, using occupation-month observations. Occupation fixed effects absorb permanent differences in requirements across occupations, and calendar-month fixed effects account for seasonality. We also control for average posting length, since longer advertisements may list more skills. We find an annual increase of 1% in soft skill demand, and less than 1% in technical skill demand. This sums up to an overall increase of 16% in soft skill demand and 6% in technical skill demand over the period 2011-2024.

The positive average trend reflects increases across most occupations. Figure 4 compares average requirements in 2011–2015 and 2016–2024 for the six-digit SOC occupations observed in both periods. Soft-skill requirements rise in 88% of occupations, accounting for 90%

Figure 3: Time Series For Skill Demand



Notes: This figure shows changes in average skill counts per posting in Lightcast from 2011–2024. Soft skills follow the taxonomy rules in Appendix A1.2; technical skills comprise the remaining labels, including certifications. Monthly averages are total skill mentions divided by total postings, calculated separately for each category. Each series is smoothed using an HP filter with $\lambda = 14,400$ and then normalized to 100 in January 2011. These series use skill-label counts; the separate LLM intensity measure rates the requirements described in full advertisements.

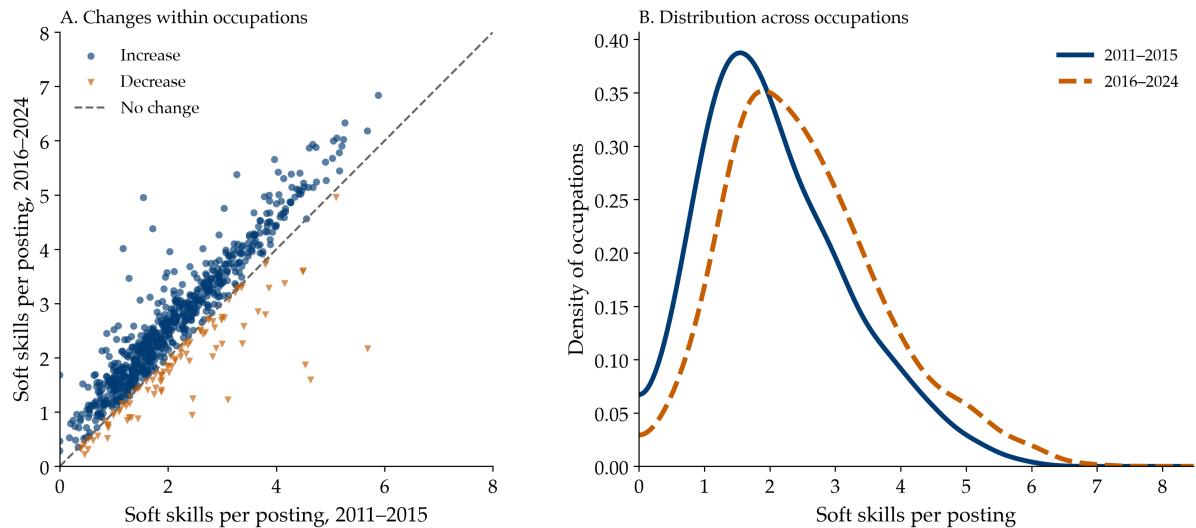
Table 2: Rising Demand For Soft Skills

Dependent Variable:	Log(All Skills)	Log(Soft Skills)	Log(Technical Skills)
% change per year	0.60*** (0.12)	1.20*** (0.24)	0.48*** (0.12)
<i>Fixed-Effects:</i>			
Occupation	Yes	Yes	Yes
Month	Yes	Yes	Yes

Notes: This table shows the slope of a linear time trend for skill demand after controlling for occupation and month fixed effects (Lightcast, 2011–2024). The unit of observation is an occupation $\times$ month $\times$ year group, where occupations are classified at the 6 digit SOC code level. Soft (technical) skill demand is calculated as the average number of soft (technical) skills required in a job posting in a given occupation $\times$ month $\times$ year group. All Skills = Soft Skills + Technical Skills. Coefficients are multiplied by 12 to reflect annual changes in skill demand. We also control for the average number of words per job posting within an occupation $\times$ month $\times$ year group. Each observation is weighted by the number of job postings within the group. Standard errors are clustered at the occupation level. $* \Rightarrow p < 0.1, ** \Rightarrow p < 0.05, *** \Rightarrow p < 0.01$.

of postings in the initial period. Panel B shows the accompanying shift in the distribution, giving each occupation equal weight. Technical requirements also rise in most occupations (85%).⁶

Figure 4: The Shift in Occupational Skill Requirements



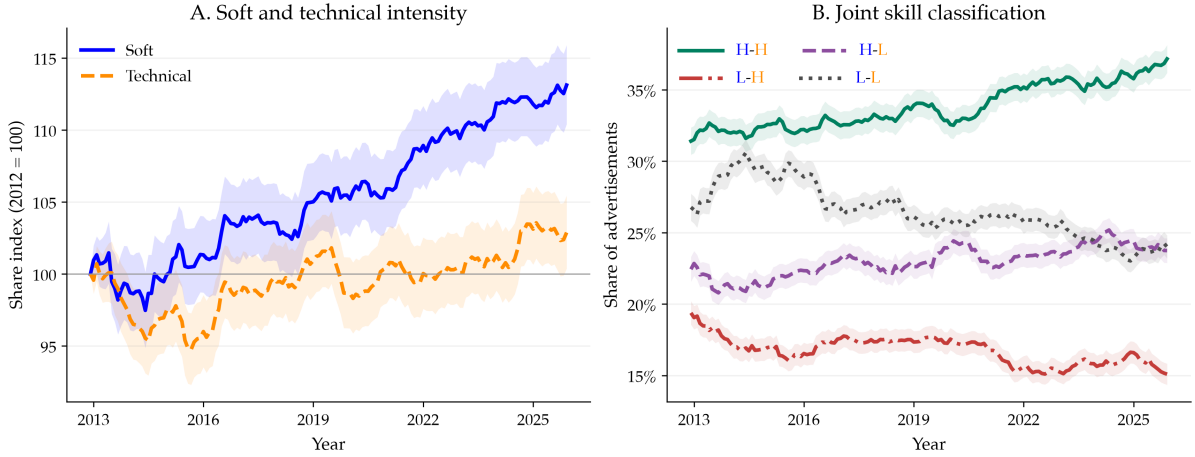
Notes: The sample consists of all six-digit SOC occupations observed in both 2011–2015 and 2016–2024 in the Lightcast data. Panel A plots each occupation’s average soft-skill requirements in 2011–2015 against those in 2016–2024. Blue circles indicate increases and orange triangles indicate decreases; the dashed diagonal marks no change. Each point represents one occupation. Counts are total skill mentions divided by total postings within occupation and period. Panel B shows the distribution of average soft-skill requirements across the same occupations, giving each occupation equal weight. The solid dark-blue curve denotes 2011–2015 and the dashed orange curve denotes 2016–2024. Counts use the taxonomy rules in Appendix A1.2.

While skill counts are the extensive margin, to understand whether the intensity of a skill in a particular job has changed, we create a measure of skill intensity directly using the content of individual job advertisements. We use a LLM to assign separate high or low ratings to soft and technical requirements, based on the duties and qualifications in each advertisement. We also train the model on the O*NET data and their classifications. A high rating requires substantive demand for the skill type; generic requests for good communication or teamwork receive a low rating. Figure 5 plots the occupation adjusted shares of postings rated high. Appendix A2 describes the sample, prompt, and construction.

From 2012 to 2025, the share rated high in soft skills rises from 54% to 61%, an increase of 7 percentage points, or 13%. The technical share rises from 51% to 52%, or 3%. The share rated high in both dimensions (H–H) rises from 31% to 37%, an increase of 6 percentage points, or 18%. Most of the increase in high soft-skill intensity therefore occurs in advertisements that also require high technical intensity. This supports the evidence from skill counts: within broad occupational groups, advertisements increasingly describe duties that place substantial demands on soft skills alongside technical expertise.

⁶Appendix Figure A2 compares growth in the two skill categories and shows the occupational map in each period. We see a rightward shift in most occupations, which suggests that the jobs are becoming more intensive in both technical and soft skills.

Figure 5: Skill Intensity in Job Advertisements



Notes: Panel A shows the soft-high and technical-high shares, each indexed to 100 for calendar year 2012. Panel B shows the four joint classification shares in levels. In its legend, the first letter is blue and denotes soft skills; the second is orange and denotes technical skills. H means high and L low. Separate line colors and patterns identify the four combinations, whose shares sum to one in each window. Both panels pool advertisements over rolling twelve-month windows and use sampling weights within each of 85 three-digit SOC groups; group shares are held fixed at their mean annual weighted shares over 2012–2025. The common sample contains 99,603 advertisements with both ratings. Shading shows approximate pointwise 95% confidence intervals. No word-count control is included.

Having documented the rise in soft-skill requirements, we combine CPS and Lightcast data to show that young workers perform relatively better in occupations that require more technical skills, but they are worse off if their job requires higher social skills. First, using Lightcast data, we calculate the average number of soft skills and technical skills required per posting in each occupation and month. For the wage regressions, we assign these occupation–month averages to CPS respondents using their occupation codes and interview month. We regress respondents’ wages on the average advertised skill requirements in their occupation and month. We interact the skill demand measures with an indicator for young workers, to understand whether young workers are differentially affected in jobs requiring higher skill levels. Formally, we estimate

$$\begin{aligned} \log w_{iot} = & \beta_s \text{Soft}_{ot} + \beta_\tau \text{Tech}_{ot} + \gamma \text{Young}_i \\ & + \delta_s (\text{Young}_i \times \text{Soft}_{ot}) + \delta_\tau (\text{Young}_i \times \text{Tech}_{ot}) \\ & + X_i' \Theta + \lambda_t + \varepsilon_{iot}, \end{aligned}$$

where i indexes CPS respondents, o occupations, and t months; Soft_{ot} and Tech_{ot} are the average soft and technical skills per posting in occupation o in month t , taken from Lightcast; Young_i is an indicator for whether respondent i is 25 years old or younger; X_i contains sex, race, and education; and λ_t are month fixed effects. The coefficients of interest are δ_s and δ_τ , which capture whether young workers are paid differentially in occupations requiring more soft or technical skills. We estimate this specification separately for college-educated and non-college-educated workers, using CPS MORG wage data from 2011–2024 for wage-employed respondents aged 55 or younger, merged with Lightcast skill demand at the occupation $\times$

month level.

Table 3 shows that the wage gap between young and older workers is larger in occupations with higher soft-skill requirements. Among college graduates, one additional soft skill per posting is associated with a 3-log-point reduction in young workers' wages relative to those of older workers. The corresponding estimate for workers without a college degree is 3 log points. The negative association is therefore present in both education groups.⁷

The relationship with technical requirements differs by education. Among college graduates, one additional technical skill per posting is associated with a 1-log-point increase in young workers' relative wages. Among workers without a college degree, it is associated with a 1-log-point decrease. In the baseline specification, young graduates thus have a smaller wage disadvantage in occupations that require more technical skills and a larger disadvantage in those that require more soft skills. The technical-skill advantage is sensitive to controlling for occupation fixed effects, however: the college interaction falls to essentially zero. The soft-skill disadvantage remains.

Table 3: Young Workers Earn Less Than Older Workers

Dependent Variable: Log(Wages)	College Educated	Non-College Educated
Young (γ)	−37.24*** (3.96)	−21.85*** (2.50)
Soft skills (β_s)	2.66* (1.55)	0.72 (1.13)
Technical skills (β_t)	2.96*** (0.31)	3.55*** (0.45)
Young $\times$ Soft (δ_s)	−3.00** (1.45)	−2.52*** (0.95)
Young $\times$ Technical (δ_t)	0.93*** (0.33)	−0.80*** (0.27)
Observations	572,091	931,070

Notes: This table shows the differential returns from soft and technical skills to young and older workers (CPS Merged Outgoing Rotation Group + Lightcast, 2011 - 2024). The wage variable reports hourly wages. For CPS respondents for whom this data is missing, hourly wages are calculated by dividing weekly wages by usual hours worked per week (if this data is available). Young = 1 if age ≤ 25 , and 0 if age $\in [26, 55]$. A respondent is college educated if she reports obtaining a bachelor's or advanced degree. The Soft (Technical) skills variable is the average number of soft (technical) skills required per job posting in an occupation $\times$ month group in Lightcast. The regression controls for standard demographic variables and time fixed effects. Standard errors are clustered at the occupation level. * $\Rightarrow p < 0.1$, ** $\Rightarrow p < 0.05$, *** $\Rightarrow p < 0.01$.

One interpretation of the soft-skill wage gap is that work experience provides skills that college alone does not supply. If these skills develop on the job, young workers may enter employment at a disadvantage even after completing college. The regressions relate wages to the requirements of workers' occupations; they do not directly measure workers' skills or establish how those skills are acquired.

Under this interpretation, the rise in soft-skill requirements could increase the importance of experience at labor market entry. Employers may become less willing to hire young graduates if the skills they require take time to acquire on the job. The wage evidence motivates

⁷ Appendix Table A7 reports alternative specifications, and Appendix Figure A14 reports estimates by year. The soft-skill interaction remains negative across the alternative specifications, although the college estimate with age controls is less precise ($p = 0.064$).

this mechanism, but does not establish its contribution to the rise in relative unemployment. We use the model to examine how a shift in skill demand changes firms' hiring and vacancy creation, and what those adjustments imply for workers entering with fewer skills.

III Model

We will now build a general equilibrium, search and matching model of the US labor market, using the model developed by [Lise and Robin \(2017\)](#) as a baseline. This model will allow for heterogeneous worker and firm types. The goal of the model is to quantify how much of the worsening labor market outcomes of young college graduates can be attributed to the rise in demand for social skills by employers.

III.A Workers, Firms, and States

Time is discrete. The economy contains a continuum of infinitely-lived workers identified by their skill level $x \in \mathcal{X} = [0, 1]$, and a continuum of firms characterized by productivity $y \in \mathcal{Y} = [0, 1]$. A worker's skill level is fixed over her lifetime, and the distribution of worker skills $\ell(x)$ is exogenous. The distribution of productivities across firms is uniform. Both workers and firms are risk neutral.

The aggregate state of the economy is given by z_t . At the beginning of each period t , the aggregate state evolves according to a Markov transition kernel $\Pi_z(z | z)$.

III.B Production and Flow Value of Unemployment

The output produced by a match between a worker with skills x and a firm with productivity y at date t is given by a second-order Taylor polynomial in the worker and firm type, scaled up by the level of aggregate shock z_t .

$$p_t(x, y, z) = z_t(a_1 + a_2x + a_3y + a_4x^2 + a_5y^2 + a_6xy) \quad (1)$$

This quadratic specification allows for the output to be non-monotonic in either the firm's or the worker's type. For example, if $a_5 < 0$, then it is possible that for some worker types x , moving to a high-type firm $y' > y$ leads to a lower level of output $p_t(x, y', z) < p_t(x, y, z)$. Similarly, the production function allows for complementarity between the worker and firm types (if $a_6 \neq 0$). Therefore, this specification will be an important driver of sorting between firms and workers, which will become apparent later.

For an unmatched worker, the flow value of unemployment is given as a fixed proportion of the output she can produce at her best possible firm, in the absence of any aggregate shocks:

$$b_t(x) = 0.7 \times p(x, y^*(x, 1), 1) \quad (2)$$

where the 0.7 factor is taken from [Hall and Milgrom \(2008\)](#), and $y^*(x, 1) = \arg \max_y p_t(x, y, 1)$ is the firm type that maximizes a worker's output in the absence of aggregate shocks ($z = 1$).

III.C Labor Markets

Consider the labor market at the beginning of period t . Let $u_t(x)$ denote the mass of unemployed workers with skills x , and let $h_t(x, y)$ denote the mass of employed workers with skills x currently matched with firm y . The stock of employed and unemployed workers should satisfy

$$u_t(x) + \int h_t(x, y) dy = \ell(x).$$

The timeline of the labor market is as follows: The period t starts with a change in the aggregate state from z_{t-1} to z_t . Separations between workers and firms happen, since some matches that were viable in period $t-1$ (i.e., generated a positive surplus) are not viable anymore. The resulting stock of unemployed agents is denoted by $u_t^+(x)$ and the remaining employed workers are denoted by $h_t^+(x, y)$. The effective labor force (the mass of workers searching for jobs) is given by

$$L_t = \int u_t^+(x) dx + s \int \int_y h_t^+(x, y) dx dy, \quad (3)$$

where $s \in [0, 1]$ is the relative search efficiency of employed workers (the search effort of unemployed workers is normalized to 1).

During the search phase, firms of type y post vacancies $v_t(y)$. The total number of vacancies in the labor market is given by $V_t = \int v_t(y) dy$. Matches form according to

$$\mathcal{M}_t(L, V) = \min\{\alpha_M L^{1/2} V^{1/2}, V_t, L_t\}, \quad (4)$$

and market tightness is defined as $\theta_t = V_t/L_t$. The job finding rate is

$$\lambda_{w,t} = \frac{\mathcal{M}(L_t, V_t)}{L_t} = \alpha_M \theta_t^{1/2}, \quad (5)$$

and the job filling rate is

$$\lambda_{f,t} = \frac{\mathcal{M}(L_t, V_t)}{V_t} = \alpha_M \theta_t^{-1/2}. \quad (6)$$

Search is random. Each period, an unemployed worker receives a job offer with probability $\lambda_{w,t}$ and an employed worker receives a job offer with probability $s\lambda_{w,t}$. Similarly, a firm posting a vacancy gets in contact with a worker with probability $\lambda_{f,t}$.

III.D Values and Surplus

III.D.1 Unemployment

Let $B_t(x)$ denote the value of unemployment for a worker with skills x . If a worker remains unemployed throughout period t , she receives the value $b_t(x)$ (as defined in Equation 2). In period $t+1$, this worker can receive a job offer with probability $\lambda_{w,t+1}$. We assume that the worker has zero bargaining power, so any firm y matched with this worker will offer her reservation value $B_{t+1}(x)$. If the worker does not receive a job offer (with probability $1 - \lambda_{w,t+1}$), she remains unemployed and receives her value of unemployment $B_{t+1}(x)$. Therefore, the expression for $B_t(x)$ is given as follows:

$$B_t(x) = b_t(x) + \beta \mathbb{E}_t \left[(1 - \lambda_{w,t+1}) B_{t+1}(x) + \lambda_{w,t+1} \int \frac{v_{t+1}(y)}{V_{t+1}} B_{t+1}(x) dy \right] \quad (7)$$

$$= b_t(x) + \beta \mathbb{E}_t B_{t+1}(x) \quad (8)$$

where β is the discount factor and $\mathbb{E}_t$ is the expectation taken at time t with respect to the future aggregate states z .

III.D.2 Surplus

Let $P_t(x, y, z)$ denote the joint value of a match between a worker x and a type- y firm. Define match surplus as

$$S_t(x, y, z) = P_t(x, y, z) - B_t(x). \quad (9)$$

A match between a worker x and firm y produces output $p_t(x, y, z)$ in period t . In period $t + 1$, after the realization of the aggregate shock z_{t+1} , the worker and firm assess whether to continue with the match or not. Since matches are efficient, a match is destroyed if $P_{t+1}(x, y, z) < B_{t+1}(x) \Leftrightarrow S_{t+1}(x, y, z) < 0$ ⁸, otherwise the match remains viable (i.e., if $P_{t+1}(x, y, z) \geq B_{t+1}(x) \Leftrightarrow S_{t+1}(x, y, z) \geq 0$). Matches can also be destroyed with an exogenous probability δ in each period.⁹

If a match remains viable in period $t + 1$, the worker draws a job offer with probability $s\lambda_{w,t}$, and the probability that this job offer comes from a firm of type y' is $v_{t+1}(y')/V_{t+1}$. Whether the worker accepts or rejects the offer depends on the value offered by its current firm $W_{t+1}(x, y, z)$ and the competing firm $W_{t+1}(x, y', z)$. We use the sequential auction framework of Postel-Vinay and Robin (2002) to model the competition between firms and determine the wages offered to the worker.

Suppose the worker's current wages at firm y are $W_t(x, y, z) \leq P_t(x, y, z)$. If $P_{t+1}(x, y', z) > P_{t+1}(x, y, z) \Leftrightarrow S_{t+1}(x, y', z) > S_{t+1}(x, y, z)$, then the two firms compete with each other by raising their wage offers, until the wage offer reaches $W_{t+1}(x, y', z) = W_{t+1}(x, y, z) = P_{t+1}(x, y, z)$ and the worker switches to the new firm y' .¹⁰ On the other hand, if $P_{t+1}(x, y', z) \leq P_{t+1}(x, y, z) \Leftrightarrow S_{t+1}(x, y', z) \leq S_{t+1}(x, y, z)$, then the worker stays at its current firm, but it will receive a new wage $W_{t+1}(x, y, z) = \max[W_t(x, y, z), P_{t+1}(x, y', z)]$. Therefore, we get the following expression:

⁸the continuation value of a match post-separation is $B_{t+1}(x)$, as the firm receives a value of 0

⁹Abusing the notation slightly, we remove the time subscript on the aggregate shock z . It should be interpreted as follows: If the value function is for the period t , for example $P_t(x, y, z)$, then $z = z_t$. If the value function is for period $t + 1$, for example $P_{t+1}(x, y, z)$ then $z = z_{t+1}$. Note that a worker's type x and a firm's type y remain fixed across periods.

¹⁰At this point, firm y cannot raise its wage offer anymore, but firm y' can still raise its offer by an ϵ amount and poach the worker, so we break ties in favour of the firm which has a higher surplus with the worker.

$$\begin{aligned}
P_t(x, y, z) &= p(x, y, z_t) \\
&+ \beta \mathbb{E}_t \left[\left(\delta + (1 - \delta) \mathbf{1}\{P_{t+1}(x, y, z) < B_{t+1}(x)\} \right) B_{t+1}(x) \right. \\
&\quad \left. + (1 - \delta) \mathbf{1}\{P_{t+1}(x, y, z) \geq B_{t+1}(x)\} \left((1 - s\lambda_{w,t+1}) P_{t+1}(x, y, z) \right. \right. \\
&\quad \left. \left. + s\lambda_{w,t+1} \int \max\{P_{t+1}(x, y, z), W_{t+1}(x, y', z)\} \frac{v_{t+1}(y')}{V_{t+1}} dy' \right) \right] \quad (10)
\end{aligned}$$

$$\begin{aligned}
&= p(x, y, z_t) \\
&+ \beta \mathbb{E}_t \left[\left(1 - (1 - \delta) \mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} \right) B_{t+1}(x) \right. \\
&\quad \left. + (1 - \delta) \mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} P_{t+1}(x, y, z) \right]. \quad (11)
\end{aligned}$$

Note that $P_t(x, y, z)$ is independent of the identity of the competing firm y' . This is because if the worker remains at firm y in period $t + 1$, the output $P_{t+1}(x, y, z)$ is not affected by the competing offer (only the share received by the worker changes). If the worker switches, the firm y receives 0, while the worker receives a value $W_{t+1}(x, y', z) = P_{t+1}(x, y, z)$.

Combining equations (8), (9), and (11), we get the following expression for the match surplus:

$$S_t(x, y, z) = p_t(x, y, z) - b_t(x) + \beta(1 - \delta) \mathbb{E}_t \max\{S_{t+1}(x, y, z), 0\} \quad (12)$$

Proposition 1 in [Lise and Robin \(2017\)](#) states that the match surplus depends on time only through the aggregate productivity shock, and it does not depend on the distributions of vacancies, unemployed workers, or worker-firm matches.

III.E Vacancy Posting

There is no fixed cost required for the creation of vacancies. The cost of posting $v_t(y)$ vacancies is increasing and convex:

$$c(v_t(y)) = \kappa_0 \frac{v_t(y)^{1+\kappa_1}}{1+\kappa_1}, \quad \kappa_0 > 0, \kappa_1 > 0. \quad (13)$$

Let $J_t(y)$ denote the expected value of filling a vacancy. If a vacancy does not lead to a viable match, the firm receives 0 value. If a match is formed, the value to the firm depends on the surplus generated from the match:

$$\begin{aligned}
J_t(y) &= \int_{\mathcal{X}} \frac{u_t^+(x)}{L_t} \max\{S_t(x, y, z), 0\} dx \\
&\quad + \int_{\mathcal{X}} \int_{\mathcal{Y}} \frac{s h_t^+(x, y')}{L_t} \max\{S_t(x, y, z) - S_t(x, y', z), 0\} dy' dx. \quad (14)
\end{aligned}$$

The firm's vacancy-posting condition is

$$\kappa_0 v_t^{\kappa_1}(y) = \lambda_{f,t} J_t(y), \quad (15)$$

which gives us the following expression for the number of vacancies created by each firm type y :

$$v_t(y) = \left(\lambda_{f,t} \frac{J_t(y)}{\kappa_0} \right)^{\frac{1}{\kappa_1}}. \quad (16)$$

Substituting for the job filling rate and aggregating over firm types gives us the equilibrium tightness:

$$\theta_t = \left(\frac{1}{L_t} \int \left(\frac{\alpha_M J_t(y)}{\kappa_0} \right)^{\frac{1}{\kappa_1}} dy \right)^{\frac{\kappa_1}{\kappa_1 + 0.5}}. \quad (17)$$

III.F Worker Flows

During the separations stage, some employed workers will join the pool of unemployed either due to the exogenous job destruction shock δ , or because their match is not viable anymore ($S_t(x, y, z) < 0$). The resulting stocks of unemployed and employed workers is:

$$u_t^+(x) = u_t(x) + \int_y \left(\delta + (1 - \delta) \mathbf{1}\{S_t(x, y, z) < 0\} \right) h_t(x, y) dy \quad (18)$$

and

$$h_t^+(x, y) = (1 - \delta) \mathbf{1}\{S_t(x, y, z) \geq 0\} h_t(x, y). \quad (19)$$

During the matching and poaching stage, unemployed workers receive a job offer from a firm y with probability $\lambda_{w,t} \frac{v_t(y)}{V_t}$ and form a match with the firm if it is viable ($S_t(x, y, z) \geq 0$). The stock of employed workers at a firm come from three sources: (i) workers who were not poached by other firms, (ii) workers poached from another firm, and (iii) workers hired from unemployment. The resulting distribution of unemployed and employed workers at the end of period t (or at the beginning of period $t + 1$) is given by:

$$u_{t+1}(x) = u_t^+(x) - u_t^+(x) \lambda_{w,t} \int \frac{v_t(y)}{V_t} \mathbf{1}\{S_t(x, y, z) \geq 0\} dy \quad (20)$$

and

$$\begin{aligned} h_{t+1}(x, y) = & h_t^+(x, y) \left[1 - \int s \lambda_{w,t} \frac{v_t(y')}{V_t} \mathbf{1}\{S_t(x, y, z) < S_t(x, y', z)\} dy' \right] \\ & + \int h_t^+(x, y') s \lambda_{w,t} \frac{v_t(y)}{V_t} \mathbf{1}\{S_t(x, y, z) > S_t(x, y', z)\} dy' \\ & + u_t^+(x) \lambda_{w,t} \frac{v_t(y)}{V_t} \mathbf{1}\{S_t(x, y, z) \geq 0\}. \end{aligned} \quad (21)$$

III.G Aggregate Quantities

Aggregate output is

$$Y_t = \int \int p_t(x, y, z) h_{t+1}(x, y) dy dx \quad (22)$$

and aggregate consumption is

$$C_t = Y_t + \int b(x) u_{t+1}(x) dx - \int \kappa_0 \frac{v_t(y)^{1+\kappa_1}}{1+\kappa_1} dy. \quad (23)$$

III.H Equilibrium

Given an initial distribution (u_0, h_0) and aggregate shock Π_z , a dynamic equilibrium is a collection

$$\{B_t, P_t, S_t, J_t, v_t, u_t, h_t, Y_t, C_t, \theta_t\}_{t \geq 0} \quad (24)$$

such that:

1. B_t , P_t , and S_t satisfy the Bellman equations (8), (11), and (12);
2. J_t is consistent with expected gains from hiring and poaching, as in (14);
3. vacancy posting satisfies the firm's first-order condition (15);
4. market tightness θ_t is determined by the matching function and by search and vacancy-creation choices;
5. unemployment and employment distributions evolve according to the worker-flow equations and the state-transition kernels;
6. goods-market feasibility holds in every period;

III.I Wages

An interesting property of our model is that we do not need wages to determine labor market flows at any stage. All vacancy posting decisions and worker moves are decided based on the surplus of a match (Lise and Robin, 2017). However, we are also interested in tracking wages of workers, so we derive the expression for a worker's wages as described below.

Let σ_t be the share of surplus $S_t(x, y, z) = P_t(x, y, z) - B_t(x)$ that a worker x receives when matched with a firm y in period t . This share can depend on the history of past offers this worker has received. As in Lise and Postel-Vinay (2020), we assume that σ_t remains constant until the match dissolves or until an outside offer arrives and renegotiation happens:

$$\sigma_t(x, y, z) = \frac{W_t(x, y, z, \sigma) - B_t(x)}{P_t(x, y, z) - B_t(x)}. \quad (25)$$

Let $W_t(x, y, z, \sigma)$ denote the value a worker obtains from being employed, and let $w_t(x, y, z, \sigma)$

denote the wage that delivers this value. Then

$$\begin{aligned}
W_t(x, y, z, \sigma) = & w_t(x, y, z, \sigma) \\
& + \beta \mathbb{E} \left[\left(\delta + (1 - \delta) \mathbf{1}\{S_{t+1}(x, y, z) < 0\} \right) B_{t+1}(x) \right] \\
& + \beta(1 - \delta) \mathbb{E} \left[\mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} (1 - s\lambda_{w,t+1}(y)) W_{t+1}(x, y, z, \sigma) \right] \\
& + \beta(1 - \delta) \mathbb{E} \left[\mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} s\lambda_{w,t+1}(y) \right. \\
& \quad \times \left. \int \min\{P_{t+1}(x, y, z), \max\{W_{t+1}(x, y, z, \sigma), P_{t+1}(x, y', z)\}\} \frac{v_{t+1}(y')}{V_{t+1}} dy' \right]. \tag{26}
\end{aligned}$$

The first term denotes current period wages. The second term considers the situation where the worker becomes unemployed next period.¹¹ The third term refers to the case when the match is not destroyed and the worker does not receive a job offer from a competing firm. In this case, she receives the same share of surplus σ that she received in the previous period.

The final term provides the worker's value if an outside offer arrives. The actual value received by the worker depends on multiple cases: (i) If the outside offer is low enough, i.e. $W_{t+1}(x, y, z, \sigma) \geq P_{t+1}(x, y', z)$, where $P_{t+1}(x, y', z)$ is the maximum value that firm y' can offer, then the worker stays at her current firm and receives $W_{t+1}(x, y, z, \sigma)$, (ii) If the outside offer is large enough, i.e. $P_{t+1}(x, y', z) > P_{t+1}(x, y, z)$, then the worker switches to firm y' and receives a value of $P_{t+1}(x, y, z)$, and (iii) in the intermediate case when $P_{t+1}(x, y, z) \geq P_{t+1}(x, y', z) > W_{t+1}(x, y, z, \sigma)$, then the worker renegotiates her value up to $P_{t+1}(x, y', z)$ and stays at her current firm y .

The expression for $w_t(x, y, z, \sigma)$ can then be written by rearranging Equation (26), using Equation (25) to obtain $W_{t+1}(x, y, z, \sigma)$ in terms of $P_{t+1}(x, y, z)$ and $B_{t+1}(x)$, and then substituting the expressions for $P_{t+1}(x, y, z)$ and $B_{t+1}(x)$ using Equations (11) and (8) respectively.

$$\begin{aligned}
w_t(x, y, z, \sigma) = & \sigma_t p_t(x, y, z) + (1 - \sigma_t) b_t(x) \\
& - \beta(1 - \delta) \mathbb{E} \left[s\lambda_{w,t+1} \mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} \Omega_{t+1}(x, y, z, \sigma) \right], \tag{27}
\end{aligned}$$

where the renegotiation term is

$$\Omega_t(x, y, z, \sigma) = \int \left[\min\{S_t(x, y, z), \max\{S_t(x, y', z), \sigma_t S_t(x, y, z)\}\} - \sigma_t S_t(x, y, z) \right] \frac{v(y')}{V} dy'. \tag{28}$$

The expression inside the integral in $\Omega_t(x, y, z, \sigma)$ can be interpreted as follows:¹²

¹¹Note that even if the worker receives a job after becoming unemployed next period, her value remains $B_{t+1}(x)$ because the worker has zero bargaining power and only receives wage offers equal to her reservation value.

¹²It can also be written as $\min\{(1 - \sigma_t)S(x, y, z), \max\{0, S(x, y', z) - \sigma_t S(x, y, z)\}\}$.

$$\min\{S, \max\{S', \sigma_t S\}\} - \sigma_t S = \begin{cases} 0, & S' \leq \sigma_t S \quad (\text{offer below current wage}) \\ S' - \sigma_t S, & \sigma_t S < S' \leq S \quad (\text{worker stays, renegotiates up}) \\ (1 - \sigma_t)S, & S' > S \quad (\text{worker leaves, firm loses its full share}) \end{cases}$$

In Equation (27), the first term denotes the share of surplus that the firm pays to the worker, over and above the worker's flow value of unemployment $b_t(x)$. The second term says that the firm anticipates outside offers in the future that might lead to wage increases and/or job switches, and it compensates for this possibility by lowering the wages offered in the current period.

IV Parametrization

To understand the impact of an increase in soft skill demand on labor market outcomes, we will first fit the model to key aggregate moments for the US labor market between 1951 and 2012. In this section, we document how well our model fits the data, and the parameter estimates we obtain.

We will target the means and standard deviations of labor market flows, such as the unemployment rate (U), and the transitions from unemployment to employment (UE), employment to unemployment (EU), and job-to-job (EE). We will also target moments related to the unemployment duration profile, i.e., the number of people unemployed for longer than 5 weeks (U^{5p}), longer than 15 weeks (U^{15p}), and longer than 27 weeks (U^{27p}). Aggregate time series data on most of the above moments are obtained from BLS for our time period in consideration, except for job-to-job transitions, for which we rely on CPS data between 1994 and 2012 (after applying the corrections proposed by Fujita et al. (2024)).

For labor demand side moments, we utilize data on the standard deviation of vacancies created (V), and the standard deviation and autocorrelation of firm value added (VA). Data on vacancies is obtained from combining the BLS Help Wanted Index (HWI) for the period 1951-2006, and JOLTS for the period 2001-2012, where we project the HWI data on JOLTS data for the overlapping years to get a consistent time series. Time series data on firm value added is obtained through BLS. Additionally, we will use Compustat data to calculate the mean and standard deviation of the cross-sectional dispersion in firm value added (Bloom et al., 2018).

We also construct moments for the mean and standard deviation of labor market tightness (V/U), and for the correlations between value added and the multiple labor market variables (U , V , UE , EU). Finally, we also utilize data on the correlation between V and U , between UE and EE , and between VA and the cross-sectional dispersion in firm value added. The values for our data moments and the corresponding model moments are given in Table 4.

The model is solved at a weekly frequency, and simulated model moments are aggregated to the monthly or quarterly level to match the frequency at which the data moments are available.¹³ We set the discount rate to an annual rate of 5%, and matching elasticity ω is fixed at

¹³The reason why we choose a weekly model frequency is to reduce time aggregation bias. For example, if we

Table 4: Data Moments and Model Simulated Moments

Fitted moment	Data	Model	Fitted moment	Data	Model
$E[U]$	0.058	0.057	$sd[U]$	0.191	0.200
$E[U^{5p}]$	0.035	0.031	$sd[U^{5p}]$	0.281	0.311
$E[U^{15p}]$	0.018	0.017	$sd[U^{15p}]$	0.395	0.403
$E[U^{27p}]$	0.010	0.010	$sd[U^{27p}]$	0.478	0.420
$E[UE]$	0.421	0.474	$sd[UE]$	0.127	0.121
$E[EU]$	0.025	0.028	$sd[EU]$	0.100	0.094
$E[EE]$	0.025	0.025	$sd[EE]$	0.095	0.110
$E[V/U]$	0.634	0.756	$sd[V/U]$	0.381	0.303
$E[sd \text{ labor prod}]$	0.494	0.432	$sd[sd \text{ labor prod}]$	0.039	0.041
$sd[V]$	0.206	0.104	$corr[V, VA]$	0.721	0.999
$sd[VA]$	0.033	0.034	$corr[UE, VA]$	0.878	0.991
$autocorr[VA]$	0.931	0.994	$corr[EU, VA]$	-0.717	-0.960
$corr[V, U]$	-0.846	-0.988	$corr[UE, EE]$	0.695	0.987
$corr[U, VA]$	-0.860	-0.992	$corr[sd \text{ labor prod}, \Delta VA]$	-0.366	-0.275

Notes: U^{5p} , U^{15p} and U^{27p} are the fractions of the labor force unemployed more than 5, 15 and 27 weeks; UE , EU and EE are the monthly unemployment-to-employment, employment-to-unemployment and job-to-job transition rates; VA is aggregate value added, ΔVA its annual growth rate, and “sd labor prod” the cross-sectional standard deviation of log value added per job. The model is solved and simulated at a weekly frequency on a $21 \times 21 \times 51$ grid over (x, y, z) for 700 years, the first 100 of which are discarded, and the simulated series are then aggregated to the quarterly frequency (annual for the productivity-dispersion moments) exactly as in the data.

Table 5: Parameter Estimates

Estimate			Estimate		
Matching	α	0.4957	Worker heterogeneity	β_1	2.1454
$\alpha L^\omega V^{1-\omega}$			Beta(β_1, β_2)	β_2	11.9311
Search intensity	s	0.0274			
Vacancy costs	c_0	0.0842	Value added	p_1	0.0025
$\frac{c_0}{1+c_1} v(y)^{1+c_1}$	$1 + c_1$	1.0288	$p(x, y, z) = z(p_1 + p_2x + p_3y$	p_2	2.0457
Exogenous separation	δ	0.0127	$+ p_4x^2 + p_5y^2 + p_6xy)$	p_3	-0.1393
Productivity shocks	σ	0.0657		p_4	8.0491
Gaussian copula (σ, ρ)	ρ	0.9997		p_5	-1.9195
				p_6	6.5988

Notes: This table shows the estimated parameter values from the simulated method of moments exercise. The discount rate is fixed at 5% per annum. The model is estimated at a weekly frequency. ω is fixed at 0.5.

0.5. The distribution of workers follows a beta distribution. The worker types x and firm types y are continuous on $[0,1]$, and they are each approximated using a linearly spaced grid having 21 points. The aggregate productivity shock z follows an AR(1) process, set up in the model using the discretized Rouwenhorst method, where $\text{mean}(z_t) = 1$.

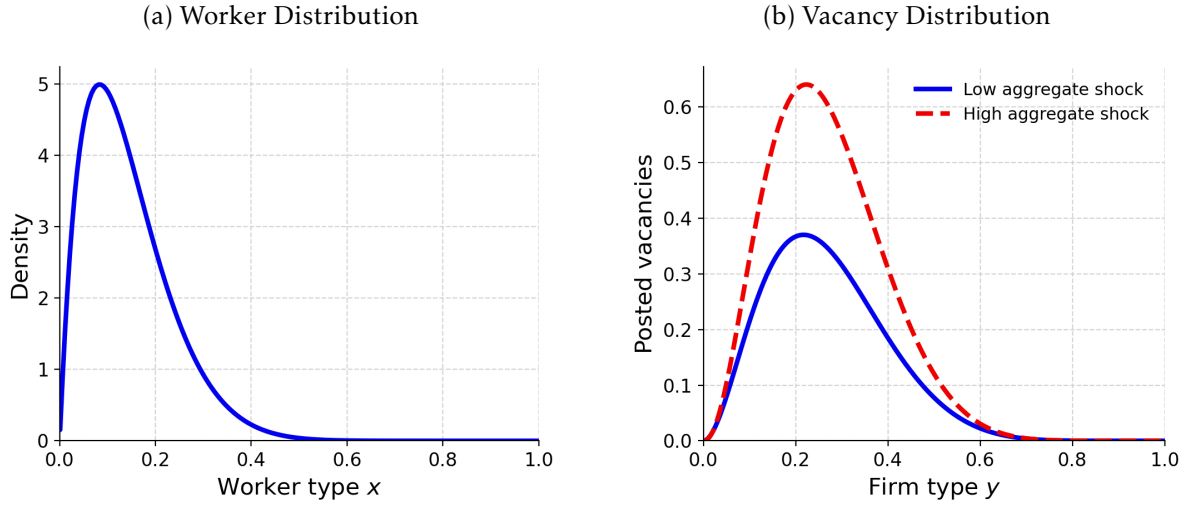
To match the model to data moments, we use simulated method of moments (SMM). The SMM objective function is nonsmooth and has many local minima, so we use a mix of global search (using differential evolution) and local refinement (using Nelder-Mead) to obtain the set of parameter estimates that minimize the objective function globally. We simulate the model for 700 years (which includes an initial burn-in period of 100 years to minimize the effect of initial conditions). Table 5 shows the parameter estimates we obtain from the simulation.

Even though none of the parameters have a clear one-to-one relationship with a moment, we can provide a rough description of which set of moments have a major influence on the identification of which parameters. The parameters for matching efficiency (α), relative search efficiency of employed workers (s), and exogenous separation rate (δ) govern movements in the labor market in the model, and they are identified primarily by the moments related to labor market flows. The parameters related to the aggregate shock process (ρ, σ) are identified by the standard deviation and autocorrelation of output.

The standard deviation of vacancies and the correlation of vacancies with output help us identify the vacancy cost function parameters (κ_0, κ_1). Worker heterogeneity (denoted by the beta distribution parameters β_1 and β_2) is identified by the moments for the unemployment duration profile. The idea is that the length of unemployment can indicate a worker's skill level, because high-skilled workers should find it relatively easier to receive job offers. Firm heterogeneity is identified by the cross-sectional variation in value added per job. The re-

decided to simulate the model at a monthly or lower frequency, we would be unable to capture the labor market transitions properly for a worker who loses her job at the beginning of the month and obtains another job at the end of the month. At a monthly frequency, this would be recorded as an EE transition although in reality it is an EU and UE transition.

Figure 6: Worker Types and Vacancy Creation



Notes: Panel (a) plots the distribution of worker types generated by the estimated parameters of the model ($Beta(\beta_1, \beta_2)$), where $\beta_1 = 2.14, \beta_2 = 11.93$. Panel (b) plots the distribution of vacancies created by firm types. The low (high) aggregate shock is equal to the 10th (90th) percentile in magnitude of the shock process.

maintaining standard deviation and correlation moments help us identify the production function parameters that govern complementarities between the firm and worker type.

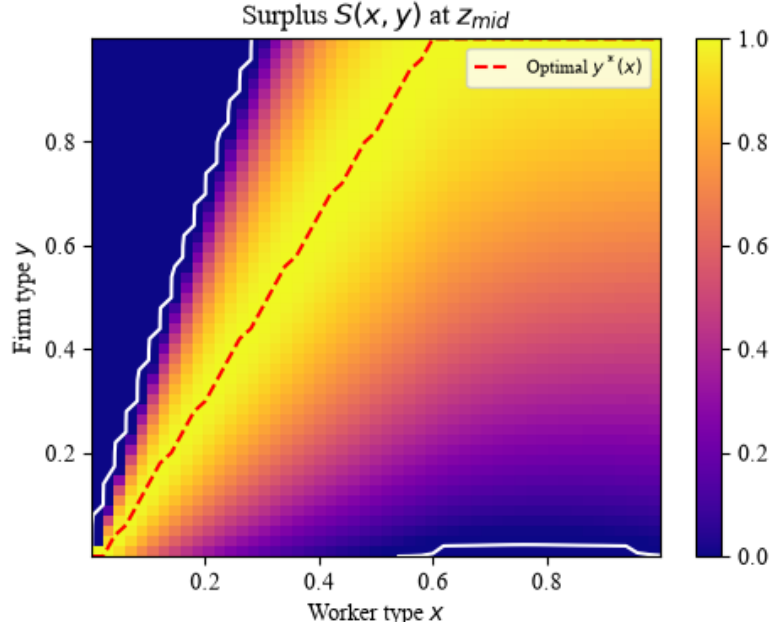
We find an aggregate matching efficiency of 0.49 at the weekly frequency. Relative search efficiency of the employed workers is 0.027, which also accounts for the fact that not all employed workers search for jobs in all periods. The small, positive value of κ_1 indicates that the vacancy cost function is slightly convex. Jobs separate at an exogenous rate of 0.013 every week. The aggregate shock process is very persistent at the weekly frequency, with $\rho = 0.999$.

The worker distribution is shown in Figure 6a. The skewness of the distribution indicates that low-skilled workers are more common in this labor market. The vacancy distribution (as shown in Figure 6b) is similar in shape to the worker distribution. This is because the firm's vacancy creation decision depends not only on the skill levels of the workers, but also on their availability. This figure also shows that if the economy is in a productive state (high aggregate shock), more vacancies will be created since a higher surplus is generated in this state.

The production function parameters give us a key insight into the functioning of the model. p_1 is small, indicating that barely any output is produced by matches between the lowest firm and worker type. Keeping the firm type fixed, production is increasing in the worker type ($p_2, p_4 > 0$). Keeping the worker type fixed, output is non-monotonic in the firm type ($p_5 < 0$). For worker types $x < 0.6$, this implies a lower output produced if they match with firms of a relatively higher type.

Figure 7 shows the surplus generated by matches between different combinations of firms and worker types, calculated at the state when $z = z_{mid} = 1$. The white lines denote the boundary beyond which surplus becomes negative. The dashed red line gives the firm type y^* that maximizes the surplus produced for each worker type x . $p_5 < 0$ implies that y^* exists (even though it might be at the corner for some worker types), and $p_6 > 0$ implies that there is positive complementarity between the firm and worker types, leading to y^* being different for

Figure 7: Match Surplus



Notes: This figure portrays the surplus generated by all possible worker-firm matches when there is no aggregate shock in the labor market ($z = z_{mid} = 1$). The white lines in the top left and bottom right areas delineate the boundary where matches create zero surplus. The dashed red line denotes the firm type $y^*(x, 1)$ which maximizes the surplus produced by a worker of type x .

different worker types.

Figure 7 indicates how sorting takes place in this labor market. For low-type workers, matches with high-type firms are unviable. Intermediate worker types continue to switch employers until they reach as close to their maximum surplus generating firm type y^* as possible. For high-type workers, surplus is increasing in the firm type, so they are most likely to match with high-type firms.

V Counterfactual

In a counterfactual exercise, we attempt to analyze how much the youth unemployment rate rises in our model when we introduce a productivity shock similar in magnitude to the rise in soft skill demand during the period 2011-2024.

The current model does not include an age dimension, nor does it have two dimensions of skills to identify soft skills and technical skills separately. Therefore, we make the following assumptions. We will assume that $x \in [0, 1]$ denotes the level of soft skills a worker has, and this is fixed over time. Consistent with our claim that young workers do not have adequate soft skills, in the model, we will classify the 10% of workers at the lower end of the skills distribution as young workers.¹⁴ Given the limitations of our current model, our current counterfactual exercise should be thought of as analyzing the impact on low-skilled workers,

¹⁴In the CPS data, young workers make up 10% of the entire labor force.

Table 6: Counterfactual Estimates

	2011	2019	2024
Data	1.91	2.33	2.57
Model	4.62	4.98	5.21

Notes: This table shows the ratio between the unemployment rates of young v.s. older college graduates in different years. Young = 1 if age ≤ 25 , and 0 if age $\in [26, 55]$. A respondent is college educated if she reports obtaining a bachelor's or advanced degree. The data values are calculated from the CPS. The 2011 model value is the unemployment rate ratio in the baseline model without the increase in labor productivity. The 2019 (2024) model value is calculated by hitting the baseline model with a productivity shock equal in magnitude to the rise in soft skill demand between 2011 and 2019 (between 2011 and 2024).

when the demand for skills increases.

We model the rise in skill demand as a productivity shock replicating skill-biased technical change (SBTC). Consider the slightly modified production function:

$$p_t(x, y, z) = z_t(p_1 + p_2(mx) + p_3y + p_4(mx)^2 + p_5y^2 + p_6(mx)y)$$

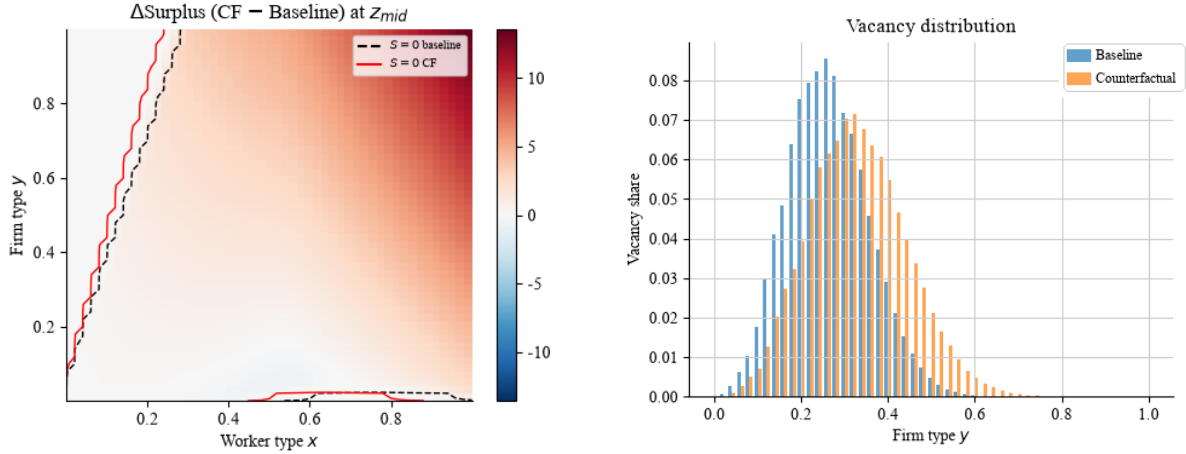
where m denotes the effective productivity of a worker of skill level x . In the baseline model, $m = 1 = m_{2011}$. SBTC is modeled as an increase in m equal to the increase in soft skill demand over 2011-2024. Therefore, we use $m_{2019} = 1.1$ and $m_{2024} = 1.16$ in the counterfactual. We refer to this technical change as skill-biased because higher-skilled workers experience a larger increase in their productivity:

$$\begin{aligned} \frac{\partial p_t(x, y, z)}{\partial m} &= z_t(p_2x + 2p_4x^2 + p_6xy) \\ \Rightarrow \frac{\partial}{\partial x} \left(\frac{\partial p_t(x, y, z)}{\partial m} \right) &= z_t(p_2 + 4p_4x + p_6y) \\ &> 0 \end{aligned}$$

Our object of interest is the ratio between the unemployment rate of young workers and the unemployment rate of old workers. We calculate this both in the data and in the model, for years 2011, 2019, and 2024 and report them in [Table 6](#). Comparing the values, we find that the unemployment rate ratio in the model increases 36% as much as the increase in the same object in the data. Therefore, we conclude that our model is able to explain 36% of the rise in youth unemployment even during the pre-COVID period 2011-2019.

Why does an SBTC shock lead to a relative rise in youth unemployment? This can be answered using [Figure 8](#). The SBTC shock makes all worker types more productive, so the set of viable matches (matches with a surplus ≥ 0) weakly expands for all workers. Firms also create more vacancies as more surplus is generated by the same worker type in a match. However, due to a larger increase in the productivity of higher-skilled workers, firms find it optimal to generate more vacancies for higher-type workers, leading to a shift in the vacancy distribution towards high-type workers. [Figure 9](#) shows the result. Even though the set of viable matches has increased for low type workers, there are fewer vacancies available for them. Their job finding rate increases in absolute terms, but it decreases in relative terms, and this leads to the relative increase in their unemployment rate.

Figure 8: Counterfactual: Change in Surplus and Vacancy Creation



Notes: Panel (a) shows the change in the set of viable matches after an increase in labor productivity, when $z = z_{mid} = 1$. The dashed black lines denotes the boundaries of the original set of viable matches in the baseline model. The solid red line shows the new boundaries after labor productivity increases. Panel (b) shows the the vacancy distributions before and after the increase in labor productivity.

VI Conclusion

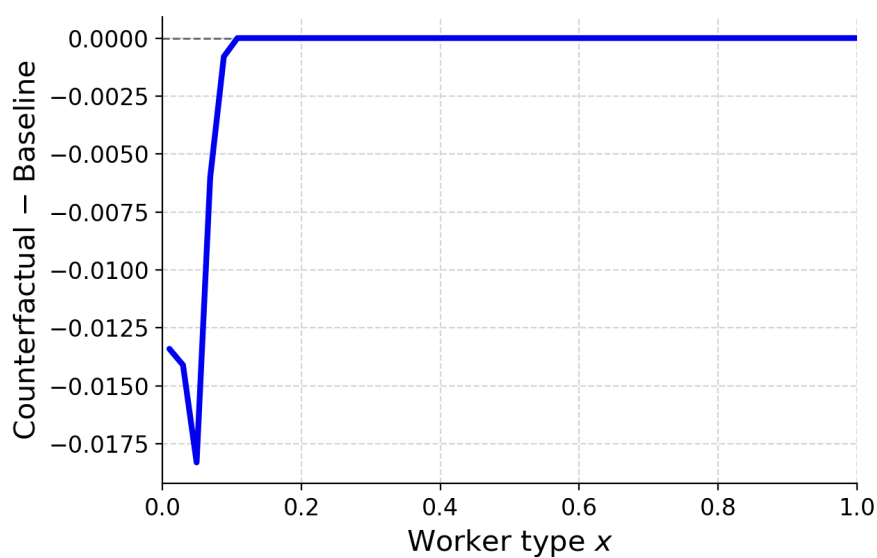
This paper set out to understand why young college graduates have fared increasingly poorly in the US labor market. We document two empirical facts. First, using near-universe data on online job postings, we show that employer demand for soft (nongognitive) skills has risen since 2011. Second, the gap between the wage of young and old workers is larger in jobs that are more soft-skill intensive. Combining both facts, we argue that as jobs become more soft-skill intensive over time, young workers would face increasingly worse labor market outcomes, as observed in the data as an increase in their unemployment rate.

We discuss that young workers fare worse in soft skill intensive jobs because soft skills are accumulated on the job, so young workers have lower levels of these skills when they enter the labor market. We build a general equilibrium, search-and-matching model, where we show that an increase in soft skill demand can account for 36% of the increase in the youth unemployment rate.

Thus far, our focus was on the unemployment rate. The rise in youth unemployment is analogous to an entry barrier to the labor market. It is possible that once young college graduates receive a job, they are able to acquire the relevant soft skills and do well in the labor market. However, similar to the related literature that finds long-run scarring effects on workers entering the labor market during a recession, it is possible that we find long-run negative effects of an increase in soft skill demand as well. Therefore, in future work, we also want to analyze the effect of soft skill demand on lifetime earnings.

Guvenen et al. (2022) show that the wages of young workers have remained stagnant since the 1990s. We also find a rising experience premium over the same time period. In our model setup, we would like to understand whether in a scenario where soft skill demand did not increase, or if young workers entered the labor market with the required amount of soft skills,

Figure 9: Fewer Viable Vacancies For Low-Skilled Workers



Notes: This figure shows the change in the share of viable vacancies available for each worker type $x \in [0, 1]$ in the counterfactual setting when labor productivity has increased. A vacancy from a firm type y is viable for a worker if the worker-firm match generates a positive amount of surplus.

their lifetime earnings would be higher or lower than what we see in the data.

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Appendix

A Empirical Analysis

A1 Data and Measurement

This appendix describes sample construction and measurement and reports the descriptive tables and figures for Section I. It covers the CPS and MORG samples, Lightcast posting characteristics, and the comparison of advertised skill counts with O*NET occupational ratings.

A1.1 CPS and MORG Samples

The descriptive sample in this appendix covers 2011–2024 and compares workers aged 18–25 with those aged 26–55. College-educated workers hold a bachelor’s degree or higher; all others are classified as noncollege. Age identifies young degree holders rather than graduation dates or actual work experience. These descriptive statistics do not describe the exact sample used in the saved wage regressions: those regressions cover ages 16–54, use real hourly earnings, and apply composition-adjusted final CPS weights. Table A7 reports the sample and weights for those estimates. The separate employment diagnostics use ages 21–29 and 30–55. These age definitions and samples need to be reconciled before the descriptive statistics serve as regression-sample summaries.

Table A1 reports descriptive statistics for 2011–2024, the common period covered by the CPS and Lightcast inputs used here. Panel A describes the basic monthly CPS population, including part-time workers, the self-employed, unemployed people, and people outside the labor force. Employment includes wage employment and self-employment; the unemployment rate divides the unemployed by the labor force. We use final CPS survey weights and exclude observations with unknown employment status from these rate denominators. The cleaned CPS file harmonizes demographic and employment categories and excludes military observations.

Panel B describes the full-time MORG wage sample with Lightcast skill requirements assigned by occupation and month. It retains wage and salary workers with at least 35 hours at their main job, positive hourly earnings, and positive ORG earnings weights. The hours measure uses usual main-job hours where available, with the substitutions documented in the CPS build. Hourly earnings are weekly earnings divided by hours, with reported hourly pay used when that ratio is unavailable. We retain allocated earnings and the existing top-code treatment. Both hourly and weekly earnings in the descriptive tables are expressed in 2024 dollars using the personal consumption expenditures price index.

This wage sample contains 1,288,565 observations. These are person-months: the CPS can interview the same person in several months. Wage statistics use ORG earnings weights, while the population statistics use final CPS survey weights. The cleaned age-restricted sample contains 9,460,549 person-months, of which 9,193,824 have observed education. The outgoing-rotation subsample contains 2,335,723 observations; requiring wage employment, at least 35 main-job hours, positive hourly earnings, and positive earnings weights leaves 1,329,640 before requiring available Lightcast requirements for the respondent’s occupation

and month. This restriction retains 96–98% of eligible earnings weight across the four age–education groups. Allocated earnings account for 32–37% of this wage sample’s earnings weight. Weekly earnings are flagged as top-coded for 2% of young college workers, 11% of older college workers, less than 1% of young noncollege workers, and 1% of older noncollege workers. These observations are retained; sample coverage alone does not establish that the wage results are insensitive to earnings allocation or top coding.

Assigning Occupational Requirements in the Wage Analysis For the wage analysis and the assigned-requirement statistics in Panel B, we map Lightcast and CPS occupation codes to common occupation groups. Within each group and month, we sum postings and skill mentions before taking their ratio. Occupations therefore contribute in proportion to their posting counts. We then merge these occupation–month averages into the CPS wage sample. Each respondent receives the average advertised requirements for her occupation and month, rather than the requirements of a particular vacancy. The posting-only exhibits below use occupation codes to group Lightcast advertisements; their construction does not require linking postings to CPS observations.

Table A1: Workers, Earnings, and Assigned Skill Requirements

	College 18–25	College 26–55	Noncollege 18–25	Noncollege 26–55
<i>A. Basic monthly CPS</i>				
Observations	261,032	2,786,779	1,528,124	4,617,889
Mean age	23.6	40.0	21.2	40.6
Women (%)	57.5	54.2	48.4	48.7
Employment/population (%)	77.9	86.9	57.3	73.6
Full-time wage employment/population (%)	61.0	71.8	31.5	56.9
Labor-force participation (%)	83.0	89.3	64.7	78.5
Unemployment rate (%)	6.2	2.7	11.4	6.1
<i>B. Full-time MORG with occupation–month skill requirements</i>				
Observations	39,294	495,521	116,755	636,995
Share with skill requirements (%)	96.3	97.9	96.5	96.3
Mean hourly earnings (2024 dollars)	28.06	43.35	17.65	25.61
Median hourly earnings (2024 dollars)	24.76	38.22	15.97	22.00
Mean weekly earnings (2024 dollars)	1,161	1,856	724	1,080
Mean weekly hours	41.4	42.9	40.8	42.0
Assigned soft requirements: mean	3.04	3.16	2.22	2.46
Assigned soft requirements: SD	1.24	1.35	1.19	1.35
Assigned technical requirements: mean	10.56	10.89	7.24	8.27
Assigned technical requirements: SD	4.32	4.38	2.98	3.52

Notes: Sample period: 2011–2024. College denotes a bachelor’s degree or higher. Observations are person-months, not unique people. Panel A uses final CPS survey weights and retains part-time workers, the self-employed, unemployed people and people outside the labor force. Employment includes wage and self-employment. Labor-market rates exclude unknown employment status; unemployment is divided by the labor force. Panel B uses ORG earnings weights and requires wage employment, at least 35 main-job hours, positive hourly earnings and available Lightcast requirements for the occupation and month. Earnings are PCE-deflated to 2024 dollars. Allocated earnings are retained and upstream top-code treatment is preserved. Assigned requirements are average skills per posting in a worker’s occupation-month, not measured worker skills. SDs describe dispersion across workers; the two panels have different samples and weights.

A1.2 Lightcast Postings and Skill Requirements

Posting Sample We measure employers' demand for skills using online job postings from Lightcast, formerly Burning Glass Technologies (Lightcast, 2024). Lightcast is a private business analytics company that collects postings from more than 160,000 sources worldwide each day. Its machine-learning algorithms extract information about the employer, location, occupation, industry, minimum experience, and posting date. For our purposes, the central feature is the list of skills requested in each posting.

The Lightcast data span January 1, 2011 through December 31, 2024. We exclude internships and postings with missing dates, occupations, industries, job titles, firm names, or skill lists. The saved inputs for the descriptive exhibits contain 307,456,240 retained postings, 769 occupation codes, and 124,480 occupation-month cells. This count has not been reconciled with the approximately 229 million postings reported in the original analysis; the available aggregates do not identify the restriction that accounts for the difference. The saved construction also excludes military occupations, Craigslist-only postings, and selected unusable employer names. A posting need not represent one position or a realized hire. The aggregate files do not retain the identifiers needed to independently check deduplication or recover attrition at each raw-data restriction.

Skill Classification The April 2025 Lightcast taxonomy contains 34,350 distinct skill labels. The confirmed soft-skill set contains 397 labels, selected using the category and named-label rules in Appendix A1.2 below.

This is a taxonomy-based count measure. It differs from classifying the text of an advertisement, which is described below. The technical category is a residual and includes more than narrowly technical expertise. The common-skills examples in Table 1 come from an earlier classification and have not yet been rebuilt under these rules. In particular, Customer Service is not in the 397-label soft-skill set; the example table and the classification statement in the main text require reconciliation.

Skill Counts and Occupational Aggregation The skill-count measures use the following rules from the April 2025 Lightcast taxonomy. Soft skills include the category Physical and Inherent Abilities except its Physical Abilities subcategory; Communication within Media and Communications; and People Management, Business Leadership, and Business Communications within Business. These rules include common and specialized skills and exclude certifications. Within Business Management, only common skills enter the soft category. We also include the individual labels Public Relations, Crisis Management, Sales, Retail Sales, Negotiation, Operations, and Consulting. All remaining labels enter the technical count, including certifications. The categories partition the distinct labels in each posting.

We measure skill demand as the average number of skills mentioned per posting, calculated separately for soft skills, technical skills, and all skills. Each distinct skill label is counted once within a posting.¹⁵ For posting j , let S_j and T_j denote the soft- and technical-skill counts.

¹⁵The measure gives equal weight to every skill mentioned in a posting. In Section A3, we examine whether O*NET can help account for differences in skill importance. The appendix shows that O*NET captures relative

For occupation o in month t , we construct

$$S_{ot} = \frac{1}{N_{ot}} \sum_{j \in (o,t)} S_j, \quad T_{ot} = \frac{1}{N_{ot}} \sum_{j \in (o,t)} T_j, \quad (29)$$

where N_{ot} is the number of retained postings. The average total skill count is $S_{ot} + T_{ot}$. A posting with an observed skill list but no soft skills contributes zero to S_{ot} ; postings without a skill list are excluded.

Table A2 reports the distributions in the saved 307,456,240-posting sample, using the taxonomy-based classification described above. Each posting receives equal weight within a period. Postings request an average of 3 soft skills and 10.2 technical skills, and 75% mention at least one soft skill. The period comparison separates 2011–2015 from 2016–2024 and retains every available year. The saved posting-count files end in December 2024.

Skill counts measure the breadth of advertised requirements. They do not measure required proficiency or workers’ own skills. Differences in the granularity of skill labels also limit comparisons between soft and technical counts: several closely related technical labels can appear in the same posting. The larger technical count therefore need not imply greater economic importance.

Coverage and Composition Table A3 reports the availability of education and experience requirements. Figure A1 shows annual posting counts. Figure 5 in the main text reports skill intensity in the separate classified-advertisement sample, using rolling twelve-month windows and a fixed three-digit occupational mix. Appendix A2 describes the text sample and scoring rules.

The occupational mix is not constant over this period. Under the fixed 2011 classification, the share of postings in soft-skill-intensive occupations falls from 74% to 58%, and the share in technical-skill-intensive occupations falls from 74% to 63%. To examine changes beyond these two groups, let p_{ot} denote occupation o ’s share of all retained postings in year t . The distance $D_t = \frac{1}{2} \sum_o |p_{ot} - p_{o,2011}|$ equals zero when the occupational distribution is unchanged. In 2024 it is 0.241 at the detailed occupation level and 0.177 after aggregation to two-digit SOC groups. These changes make it important to account for occupational composition when comparing advertised requirements over time.

skill requirements across occupations, but does not reproduce the aggregate growth in requirements observed in Lightcast. It also discusses using AI with O*NET ratings to classify the soft- and technical-skill intensity of Lightcast postings.

Table A2: Job Postings and Advertised Skill Requirements

	2011–2015	2016–2024	2011–2024
Postings (millions)	62.56	244.90	307.46
Occupations	767	767	769
Occupation-month cells	43,740	80,740	124,480
<i>Soft skills</i>			
Mean	2.84	3.04	3.00
Standard deviation	3.02	3.13	3.11
Median	2.00	2.00	2.00
90th percentile	7.00	7.00	7.00
Postings mentioning any (%)	72.88	75.36	74.86
Mean, conditional on mentioning any	3.90	4.04	4.01
<i>Technical skills</i>			
Mean	9.84	10.29	10.20
Standard deviation	9.17	8.75	8.84
Median	8.00	8.00	8.00
90th percentile	21.00	22.00	22.00
Postings mentioning any (%)	98.86	99.07	99.02
Mean, conditional on mentioning any	9.95	10.39	10.30

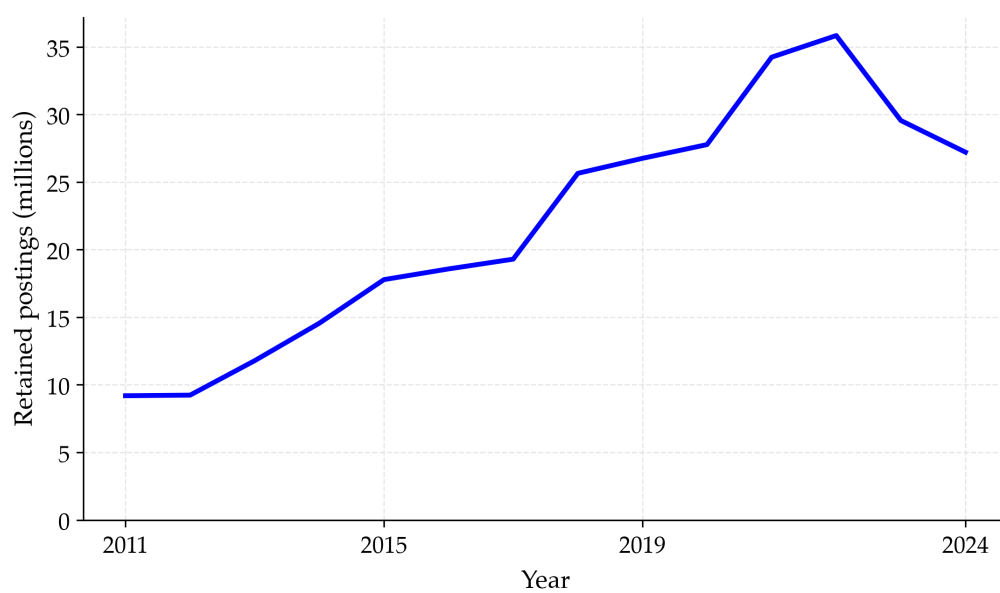
Notes: An observation is a retained posting. Each posting receives equal weight; repeated skill labels within a posting are counted once. The source excludes postings without an observed skill list, so zero means no skill in that category, not missing skills. Technical skills are the non-soft residual under the existing taxonomy-based classification. Percentiles use the inverse empirical distribution; standard deviations describe the posting distribution. Periods pool postings rather than weighting years equally. Counts need not represent unique positions or hires. Source: saved Lightcast count files, 2011–2024.

Table A3: Coverage of Education and Experience Requirements

	2011–2015	2016–2024	2011–2024
Postings (millions)	62.55	244.88	307.43
Missing education requirement (%)	44.00	45.49	45.19
Missing experience requirement (%)	51.41	53.08	52.74
Bachelor's or higher required (%)	30.99	25.48	26.60
Listed as full-time (%)	88.48	84.88	85.61

Notes: This table uses the separate education/experience-crossed source, containing 307,433,363 postings. It has 22,877 fewer postings than the main skill-count sample because the upstream aggregates use different joins. Each percentage uses this table's own posting count. Missing requirements are not coded as zero; the college-required share uses all retained postings, including those with missing education. Full-time is Lightcast's category (more than 32 hours), which differs from the CPS threshold of at least 35 hours.

Figure A1: Number of Job Advertisements

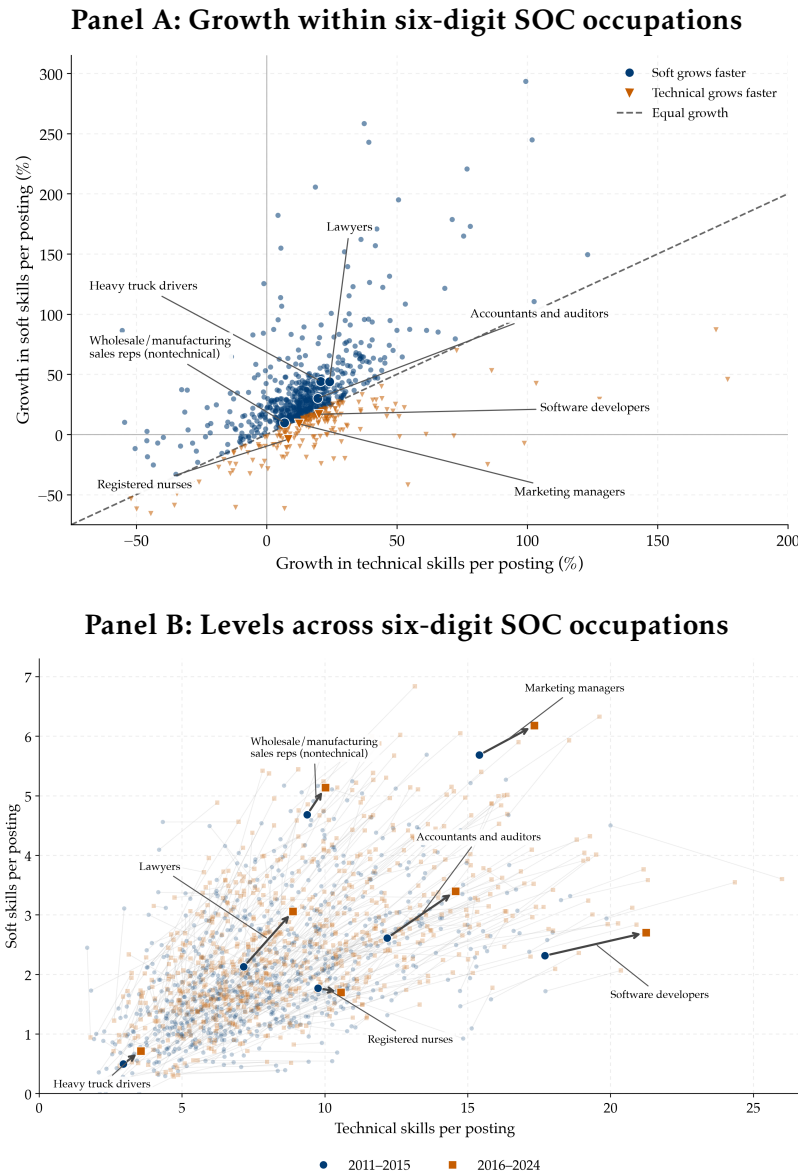


Notes: Annual counts of retained advertisements in the main Lightcast skill-count sample, 2011–2024. The sample contains 307,456,240 postings. These counts describe the full sample, rather than the subsample classified from ad text.

A1.3 Occupational Skill Requirements by Period

Figure A2 compares soft and technical requirements in 2011–2015 and 2016–2024. Panel A compares their growth within six-digit SOC occupations. Panel B shows the changes in levels for the same occupations.

Figure A2: Soft and Technical Requirements across Occupations by Period

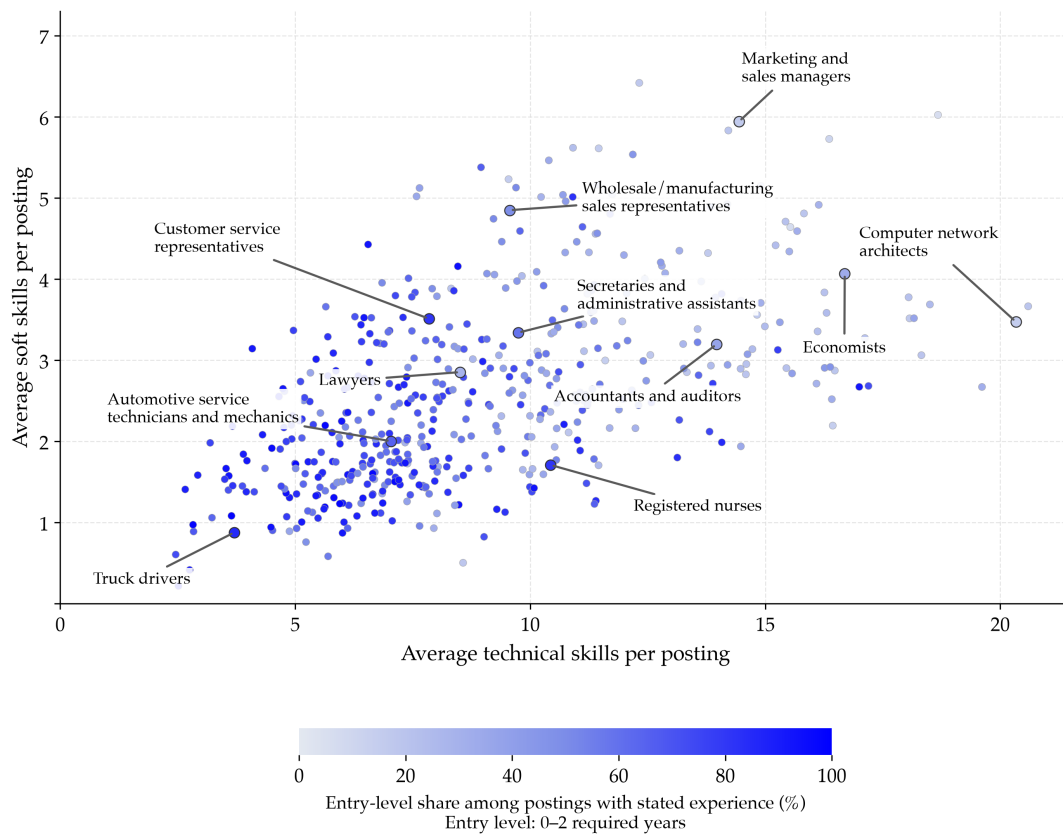


Notes: The comparison uses all 765 six-digit SOC occupations observed in both periods in the Lightcast skill-count extract. Four codes observed in only one period cannot be compared. Growth is the percentage change in mean skills per posting between the two periods, not an annualized rate. The scatter includes 762 occupations; blue circles above the dashed equality line indicate faster growth in soft skills. Three occupations with zero initial soft-skill mentions are omitted from Panel A because their percentage growth is undefined. Panel B includes all 765 occupations. Lines connect 2011–2015 (blue) to 2016–2024 (orange); arrows and darker markers highlight seven illustrative occupations, also labeled in Panel A. In both panels, means are ratios of total skill mentions to total postings within each occupation and period. Soft skills follow Appendix A1.2; technical skills comprise the remaining labels, including certifications.

A1.4 Occupational Skill Requirements and Entry-Level Postings

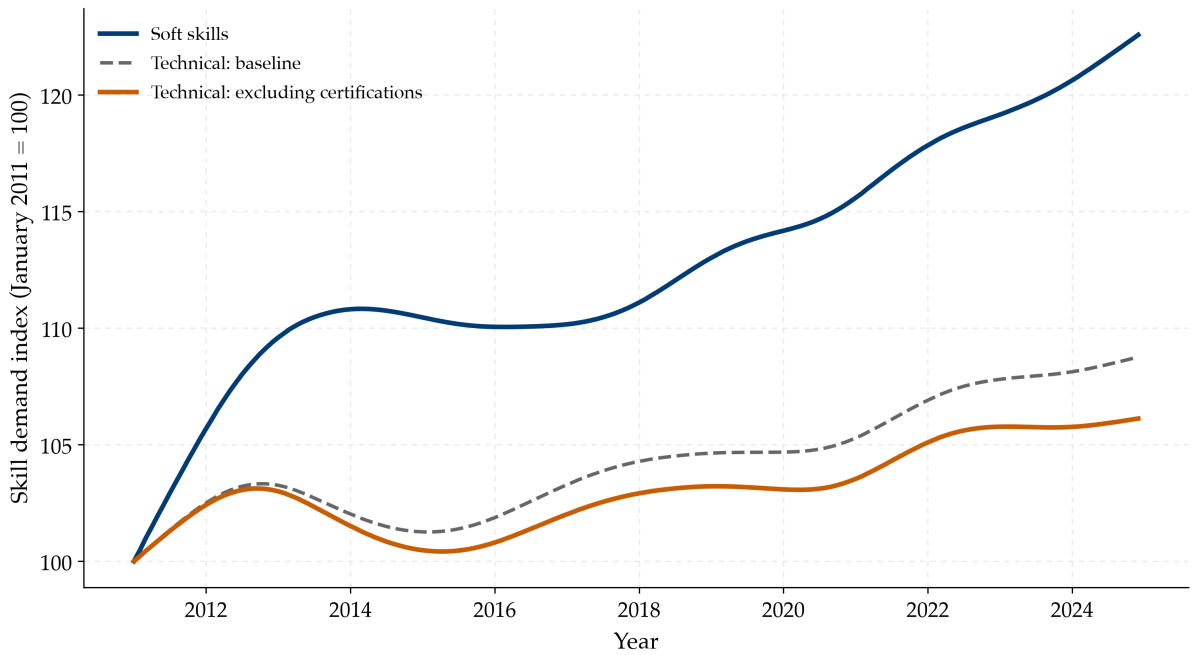
Figure A3 adds each occupation's share of entry-level postings to the skill map in Figure 1. Among postings with stated experience requirements, the entry-level share is 81% for customer service representatives and 16% for marketing and sales managers. Both occupations have relatively high soft-skill counts, but their advertised experience requirements differ sharply.

Figure A3: Occupational Skill Requirements and Entry-Level Posting Shares



Notes: Each point represents one of 494 occupation groups formed by mapping Lightcast SOC codes through the occupation crosswalk. The sample pools 297,337,527 postings from 2011–2024 across all education requirements and employment types. The axes show average technical and soft skills per posting. Soft skills follow Appendix A1.2; technical skills comprise the remaining labels, including certifications. Darker blue indicates a larger entry-level share: postings requiring a minimum of 0–2 years of experience divided by all postings with stated experience in that occupation. Posting counts are summed across years before calculating the share. Experience is stated in 141,307,614 postings; the 52% with unspecified experience are excluded from the color calculation, while the skill averages retain postings with unspecified experience. Required experience describes the advertisement, not the worker ultimately hired. Outlined points mark the labeled occupations.

Figure A4: Skill Demand Trends with and without Certifications



Notes: Monthly means divide total skill mentions by total postings in the same Lightcast sample as Figure 3. The blue curve shows soft requirements; the gray dashed curve shows baseline technical requirements; the orange curve excludes certifications from technical counts. All three series use the baseline HP smoothing parameter of 14,400 and are normalized to 100 at their respective January 2011 trend values. No postings are removed when certifications are excluded. These aggregate series allow the occupational composition of postings to change over time.

A2 LLM Classification of Job Advertisements

We use Claude Sonnet 4.5 to measure the intensity of soft and technical requirements in individual Lightcast advertisements. The model reads the advertisement and assigns an independent low or high rating to each dimension. The exercise measures how skills enter the advertised role, complementing the skill-label counts used elsewhere in the paper. Figure 5 reports our preferred series, which holds three-digit occupational composition fixed and pools advertisements over rolling twelve-month windows. Panel A reports the two marginal high-rating indices, and Panel B reports the shares of all four joint classifications in levels.

Sample. The classified sample contains 100,128 advertisements dated January 2012 through December 2025, with 596 advertisements in each month and 7,152 in each year. Both ratings are available for 100,048 advertisements; 80 observations missing both ratings are excluded. Posting identifiers are unique. We use the supplied sampling weights in all reported shares. This text sample is separate from the full posting-count sample, which covers 2011–2024.

Joint distribution. Table A4 reports the pooled shares of all four soft–technical combinations in the full valid classified sample. Figure 5, Panel B, shows their time paths in levels, using the same fixed three-digit occupational composition and rolling twelve-month windows as Panel A.

Table A4: Joint Skill Classification in the Full Advertisement Sample

Combination	Advertisements	Weighted share (%)
H-H	33,597	34
H-L	23,029	23
L-H	17,012	17
L-L	26,410	27
Total	100,048	100

Notes: The first letter denotes soft skills and the second technical skills; H means high and L low. Shares use the supplied sampling weights and pool all advertisements with both ratings over 2012–2025, across all valid occupations. The 80 advertisements missing both ratings are excluded. No occupation or time standardization is applied. Percentages are rounded independently and may not sum to 100.

Classification prompt. The prompt asks the model to consider the full advertisement, including the title, seniority, experience, education, certifications, duties, and responsibilities. It defines the two ratings as follows:

Low: “The ad contains little or no demand for the skill type, mentions it only in generic boilerplate, or requires only basic use that is not important to the role.”

High: “The skill type is important for the main duties, repeatedly emphasized, or explicitly required at an advanced level.”

Soft skills include interpersonal and workplace capabilities: communication, teamwork, leadership, people management, client interaction, negotiation, stakeholder coordination, mentoring, planning, critical thinking, judgment, and decision making. Leading a team, advising clients, resolving conflict, or coordinating stakeholders provides substantive evidence of these requirements. Generic phrases such as “good communication skills” or “team player” receive a low rating.

Technical requirements concern occupation-specific knowledge applied through specialized methods, software, tools, equipment, systems, or procedures. The prompt covers programming, quantitative analysis, engineering, clinical and scientific methods, law, accounting and finance, auditing, and skilled trades. Examples of substantive use include designing, diagnosing, coding, analyzing, troubleshooting, and operating specialized equipment. Ordinary software use, data entry, and simple standardized procedures receive low ratings.

The prompt explicitly instructs the model to score the two dimensions independently. Evidence of technical requirements cannot lower the soft-skill rating, or vice versa. An advertisement can therefore be high on both dimensions, low on both, or high on only one; the high-intensity shares need not sum to one. Benefits, generic company-culture language, application instructions, equal-opportunity statements, and legal disclaimers are ignored. The model is instructed not to infer the project’s desired result or a time trend. These are binary assessments of advertised requirements, rather than cardinal measures of worker proficiency.

O*NET context and implementation. The classifier supplies O*NET skill-importance and skill-level ratings matched to the advertisement’s occupation and posting date. Reference rows are selected when the posting date falls within their validity interval. Importance ranges from 1 to 5 and level from 0 to 7. The prompt explains that these ratings describe the occupation

generally and may help interpret ambiguous duties; occupation or O*NET information alone cannot justify a high rating. The advertisement itself must supply the evidence.

We use Claude Sonnet 4.5 with temperature set to zero. Advertisements are submitted in batches, and the prompt requests one structured response per advertisement, retaining its identifier and separate soft and technical scores (1 for low and 2 for high). It also requests confidence assessments for each dimension and low confidence for vague, short, or inconsistent advertisements. Confidence assessments are not retained in the classified file used here and do not determine the analysis sample.

Fixed occupational composition. Let H_i^k indicate a high rating for skill type k , and let w_i be the advertisement’s sampling weight. Define the joint indicator as $H_i^{HH} = H_i^{\text{soft}} H_i^{\text{technical}}$. The other joint indicators are $H_i^{HL} = H_i^{\text{soft}}(1 - H_i^{\text{technical}})$, $H_i^{LH} = (1 - H_i^{\text{soft}})H_i^{\text{technical}}$, and $H_i^{LL} = (1 - H_i^{\text{soft}})(1 - H_i^{\text{technical}})$. The four cells of the classification are H–H, H–L, L–H, and L–L, with soft status listed first. Their shares sum to one; the soft-high share equals H–H plus H–L, and the technical-high share equals H–H plus L–H. The H–H denominator is all valid advertisements in the common sample, including those rated low in either dimension. We apply the same weighting and occupation adjustment to all cells. For each three-digit SOC group g and twelve-month window ending in month t , we calculate

$$p_{gt}^k = \frac{\sum_{i \in (g, [t-11, t])} w_i H_i^k}{\sum_{i \in (g, [t-11, t])} w_i}.$$

We retain the 85 groups represented in every twelve-month window. This common sample contains 99,603 advertisements and accounts for nearly 100% of the sampling weight among advertisements with valid ratings. Let s_{gy} be group g ’s weighted share of the common sample in calendar year y . The fixed reference share is $\bar{s}_g = \frac{1}{14} \sum_{y=2012}^{2025} s_{gy}$, giving each year equal weight. The aggregate share and plotted index are

$$P_t^k = \sum_g \bar{s}_g p_{gt}^k, \quad I_t^k = 100 \frac{P_t^k}{P_{\text{Dec 2012}}^k}.$$

Thus, December 2012 represents the January–December 2012 sample, and each subsequent point advances the window by one month. We pool weighted counts within each window before calculating group shares; we do not average monthly percentages or apply an additional time-series filter. Holding the group mix fixed removes changes in the distribution across these broad occupations. It does not hold detailed jobs or employers fixed. The same formulas apply to $k \in \{HH, HL, LH, LL\}$. Figure 5, Panel B, plots the four P_t^k shares in levels. Panel A plots the marginal soft and technical indices, each using its own calendar-year 2012 share as the denominator.

Uncertainty and interpretation. The shaded bands are approximate pointwise 95% confidence intervals. Panel A’s index bands are calculated by linearizing the log index. Panel B’s share-level bands equal the adjusted share plus or minus 1.96 times its standard error. Both panels include uncertainty in the estimated reference occupation shares. Panel A additionally includes uncertainty in the 2012 baseline and retains covariance from advertisements shared

by overlapping windows. Posting months are treated as variance strata, with advertisements assumed independent within month. The available file does not identify the original sampling strata or employer clusters, so these are approximate sampling intervals. They exclude classification error and are not simultaneous bands for the entire time path. Some occupation–window cells are sparse.

The estimates describe changes in advertised requirements. They do not measure workers’ skills or identify the effect of those requirements on employment. The H–H series measures joint advertised requirements; identifying complementarity in production would require additional evidence. No independent human-rating validation is reported for this sample. O*NET agreement alone would not provide such validation because O*NET information enters the prompt.

A3 O*NET and Lightcast

Our measure of an occupation’s skill requirements (or skill demand) is calculated using Lightcast data on job postings. As we described in [Section I](#), Lightcast is able to extract the skills that each job posting asks for. We classify these skills as soft or technical skills, and we define soft skill demand at the occupation $\times$ month level as the average number of soft skills per job posting within an occupation $\times$ month group (and similarly for technical skill demand). A drawback of using skills data from job postings is that we cannot observe the importance of each skill in a particular occupation. Instead, in our skill demand measure, all skills demanded in an occupation are weighted equally.

An alternative source of information on occupation skill requirements is the Occupational Information Network ([O*NET, 2026](#)). O*NET data consists of 35 skills and approximately 900 occupations, and it assigns an importance value between 1-5 and a level between 0-7 for each skill to each occupation. Importance signifies whether a skill is required in an occupation, and the level signifies the expertise in a skill that is needed in the occupation. For example, O*NET explains that the speaking skill is important for both lawyers and paralegals. However, lawyers are required to have a higher level in this skill than paralegals.¹⁶ In this section, we will explore O*NET skills data to understand its suitability for our analysis.

O*NET updates its data on occupation skill requirements each year to capture any changes in the tasks involved in an occupation over time. We create binned scatterplots to compare how skill requirements have changed in occupations between 2011 (Version 16.0) and 2024 (Version 30.2). In [Figure A5](#) and [Figure A6](#), we see that for both the importance and level plots, the points lie close to the 45 degree line for both soft skills and technical skills, respectively. This would suggest that the O*NET occupational skill requirements remain relatively unchanged over the time period of our analysis.

Does this mean that skill demand is stable over our time period, and that our Lightcast measure of skill demand is incorrect? Here is why we think this is not the case. [Figure A7](#) shows binned scatterplots where we use our skill demand measure from Lightcast (average skills per posting in an occupation) instead of the O*NET skills requirement data. We observe

¹⁶You can find the data dictionary for the latest O*NET Version at the time of writing here ([Version 30.2](#)). You can also find O*NET’s definition of soft skills [here](#).

that there is a near uniform increase in skill demand (both soft and technical) across occupations between 2011 and 2024, irrespective of the skill demand in an occupation in 2011.

O*NET data may be unable to capture this near-uniform increase in skill demand because of 2 potential reasons. Both reasons relate to the scales used to measure the importance and level of skill requirements in O*NET. First, given that the scales are fixed (1-5 for importance and 0-7 for level), an increasing trend in skill demand would eventually lead to all skill requirements reaching the upper bound of these scales if O*NET attempted to include this information in its measures. Therefore, we believe that although O*NET data is useful for comparing skill requirements across occupations at a particular time, it is not suitable for understanding the trends in these skill requirements over time. In short, O*NET should be used for studying relative skill demand across occupations, not absolute skill demand.

Figure A5: O*NET Soft Skill Requirements (2011-2024)

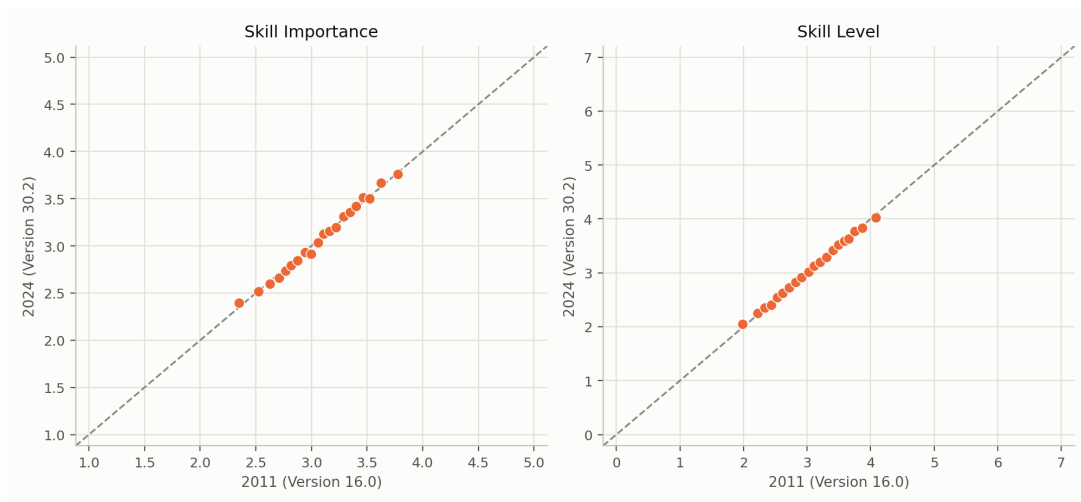
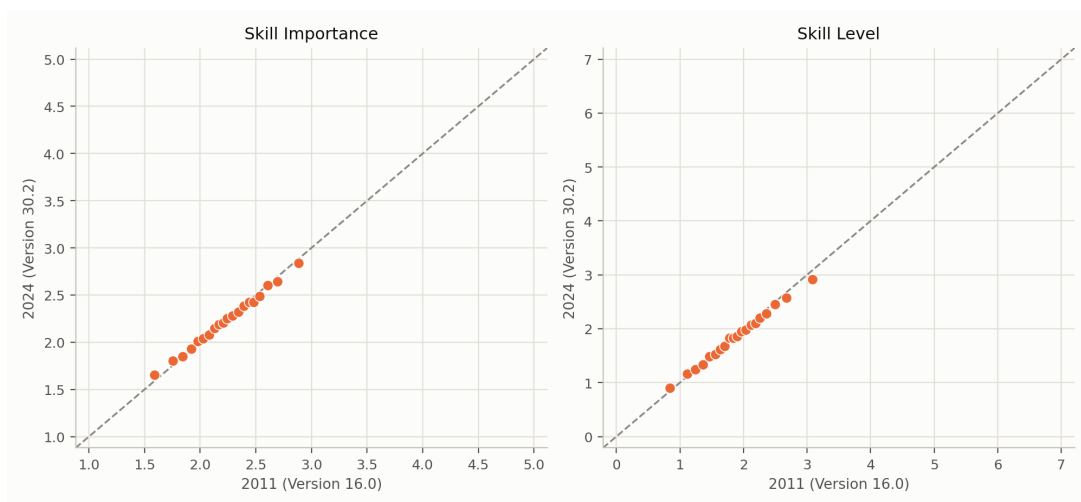


Figure A6: O*NET Technical Skill Requirements (2011-2024)

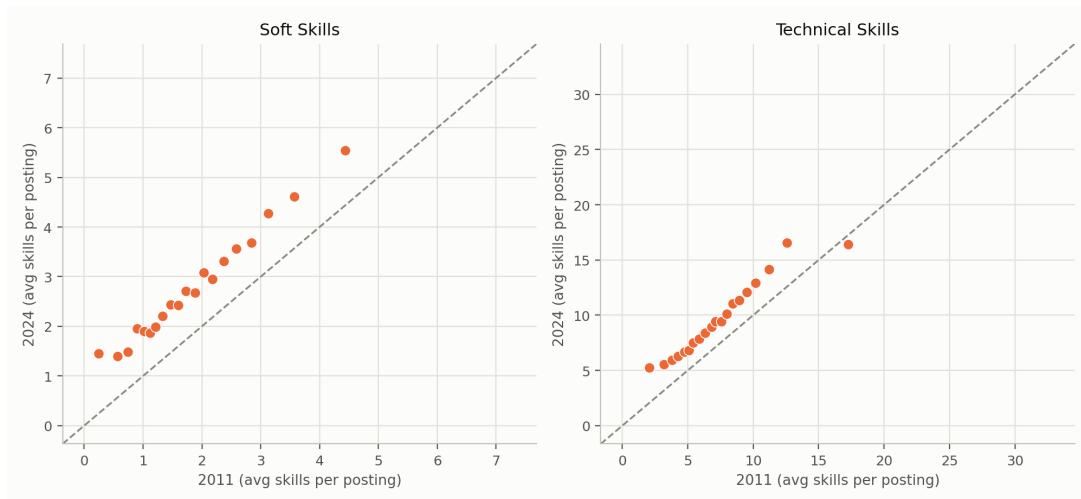


Finally, because all occupations experience a similar increase in skill demand (as observed in [Figure A7](#)), the relative skill requirements across these occupations also remains the same.

If, for example, less soft-skill intensive occupations experienced a greater increase in soft skill demand, we would have seen a flatter line formed by the points in the binned scatterplot in [Figure A5](#) relative to the 45 degree line.

To summarize, the O*NET and Lightcast data combined paint an interesting picture about the nature of the rise in skill demand. It suggests that the core tasks of an occupation remain unchanged over time (as captured by O*NET data). However, the rise in soft skill demand implies that all occupations now require an additional component, which requires workers to communicate and coordinate effectively with the people in their workplace.

Figure A7: Lightcast Skill Requirements (2011-2024)



We now show that O*NET and Lightcast capture similar information for occupations in [Table A5](#). We first classify O*NET occupations based on whether they had above median (High) or below median (Low) skill requirements in 2011. Our results are robust to using 2024 data as well. Then, we calculate the average number of skills required per job posting within these two groups using Lightcast data. [Table A5](#) shows that for different skill dimensions (soft and technical) and skill measurement definitions (importance and level), the High group requires more skills per job posting than the Low group, while the last column shows that the rank correlation between the O*NET and Lightcast skill measures is positive as well. These comparisons assess cross-sectional agreement; they do not validate changes in required proficiency over time.

To conclude, even though Lightcast data cannot measure the importance of different skills within an occupation like O*NET does, it still is able to differentiate between low and high-skill intensive occupations. Moreover, unlike O*NET, our Lightcast data is able to capture the rise in the demand for skills over time. Therefore, we choose to work with Lightcast data in our main analysis.

Cross-sectional comparison. O*NET provides information on skill requirements across US occupations, reporting both the importance of a skill and the level of proficiency required.¹⁷

¹⁷See [the O*NET overview](#) and [the scale definitions](#).

Table A5: O*NET vs. Lightcast: Construct Validity (2011)

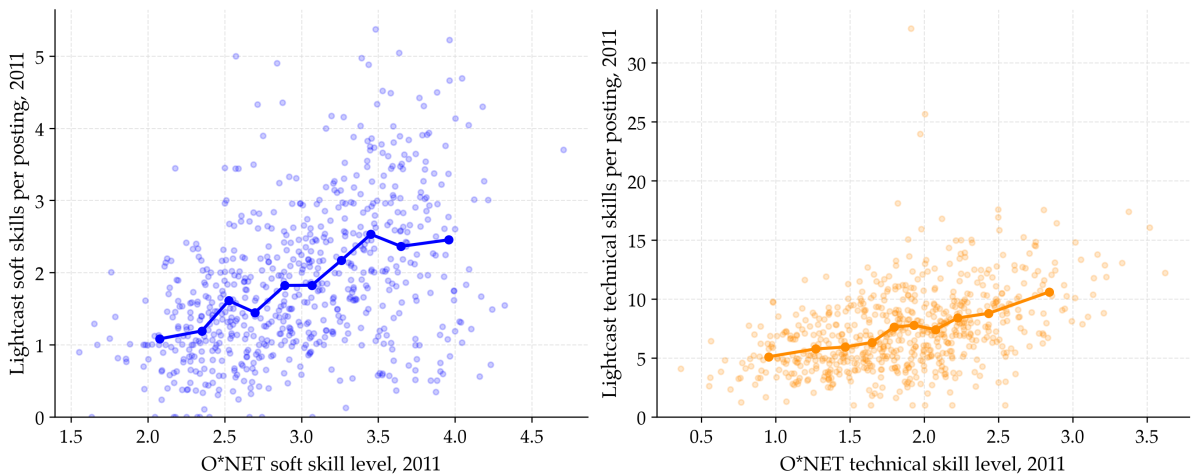
O*NET Classification	Mean, High	Mean, Low	Difference	Rank Corr. (ρ)
Soft Importance	2.30	1.43	+0.87	0.43
Soft Level	2.28	1.45	+0.83	0.45
Technical Importance	8.34	6.31	+2.03	0.39
Technical Level	8.58	6.09	+2.49	0.47

High/Low = occupations above/below the 2011 O*NET median (Importance or Level, soft or technical, as indicated). Mean columns give the group's average Lightcast avg skills per posting in 2011. Rank Corr. is the Spearman correlation between the continuous O*NET score and the continuous Lightcast level, $N = 669$. Occupations receive equal weight. The saved table does not report uncertainty for the High-vs-Low comparisons.

We use its definition of soft skills for the skill classification and its occupational ratings to assess our measures of skill demand, as described above.

Figure A8 compares the Lightcast measures with O*NET skill-level ratings in 2011 for 669 matched occupations. Occupations with higher O*NET ratings generally have higher advertised skill counts. Table A5 reports comparisons using both importance and level ratings. These comparisons assess cross-occupation differences; they do not establish that changes in counts measure changes in required proficiency over time.

Figure A8: Advertised Skill Requirements and O*NET Occupational Ratings



Notes: Each light point is an occupation in 2011. The horizontal axis is its O*NET skill-level rating; the vertical axis is the average number of skills per Lightcast posting. The connected points show means within ten quantile groups of the O*NET rating, with equal weight for each occupation. Soft and technical requirements are shown separately. The sample is the intersection of 2011 Lightcast occupations and the saved, harmonized O*NET classification panel. No causal interpretation is attached to the relationship, and the two sources measure different aspects of requirements.

A4 Additional Empirical Analysis

A4.1 Bachelor's Fields of Young College Graduates

Figure A9 shows how the bachelor's fields of young degree holders have changed since 2011. The shares with STEM and health-related degrees have risen, while the humanities, business, and social-science shares have fallen. These changes coincide with the erosion of young graduates' unemployment advantage documented in Section II; they do not directly measure work-

ers' skills or establish the cause of that erosion.

We use annual national person microdata from the American Community Survey (ACS) for 2011–2024, excluding Puerto Rico. The sample consists of civilian, noninstitutional respondents aged 18–25 with a bachelor's degree or higher. Both sexes, enrolled graduates, unemployed graduates, and those outside the labor force are included. These are repeated cross-sections of young degree holders, not a panel of students or a count of degrees awarded in each year. The age restriction does not identify the year of graduation.

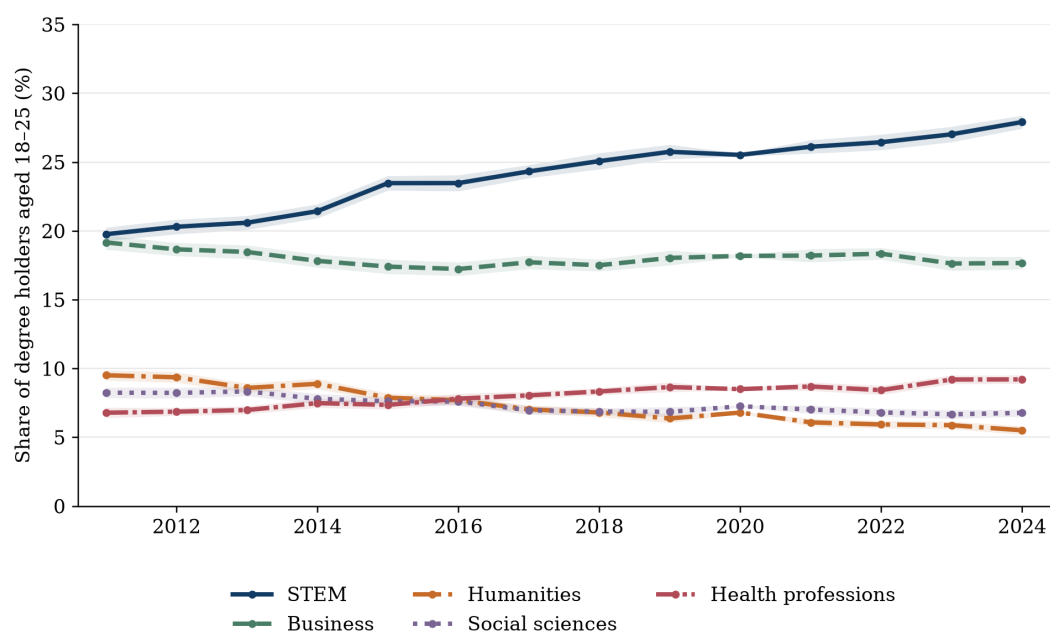
Each person counts once, using the first-listed bachelor's field (FOD1P), including for respondents with two bachelor's fields or an advanced degree. First-listed does not necessarily mean the primary field. Our core-STEM definition covers computing, engineering and engineering technologies, biology, mathematics/statistics, physical sciences and science technologies (Census prefixes 21, 24, 25, 36, 37, 50, and 51), plus mathematics/computer science, cognitive science/biopsychology, and data science/analytics (4005, 4006, and 4009). Health professions (61) is displayed separately and is not part of this STEM definition. Humanities combines area studies, languages, English, liberal arts/humanities, philosophy/religion, theology, and history (15, 26, 33, 34, 48, 49, and 64); it excludes arts. Business uses prefix 62. Social sciences uses prefix 55, including economics and sociology; psychology (52) remains outside the displayed groups. These working groups are based on the [Census field codes](#). They are not measures of workers' technical or soft skills.

For each year, a group's share is its weighted number of degree holders divided by the weighted number of all selected degree holders. Fields not displayed remain in the denominator. We use annual ACS person weights and all 80 person replicate weights; sampling variance is 4/80 times the sum of squared deviations of the replicate shares from the full-sample share. Figure bands are normal 95% confidence intervals. We include the [2020 one-year experimental release](#) using its experimental person and replicate weights. No years are interpolated and the series are not smoothed. Detailed field codes can change over time; in particular, data science has a separate code only in later releases.

Table [A6](#) reports the endpoint shares. STEM rose by 8 percentage points and health professions by 2 points. Humanities fell by 4 points, while business and social sciences each fell by about 2 points. The five displayed groups account for 63% of young degree holders in 2011 and 67% in 2024; remaining fields include education, psychology, arts, communications, and other applied fields. The graph therefore documents a shift toward STEM and health-related bachelor's fields among young degree holders, rather than an increase in their enrollment in these subjects.

Figure [A10](#) plots young graduates' unemployment rates minus the overall rate among civilians aged 16–55, across all education levels, for the same five fields and arts (Census prefix 60). Zero denotes the overall rate. Over 2011–2024, unemployment in most of the fields shown, including STEM, increased relative to the overall rate. STEM graduates moved from below the overall rate to above it, while health professions remained below it. A relative increase need not imply an absolute increase in unemployment: a field's rate can fall while the overall rate falls faster. These comparisons describe outcomes within broad fields and do not identify the cause of the changes.

Figure A9: Bachelor's Fields of Young College Graduates



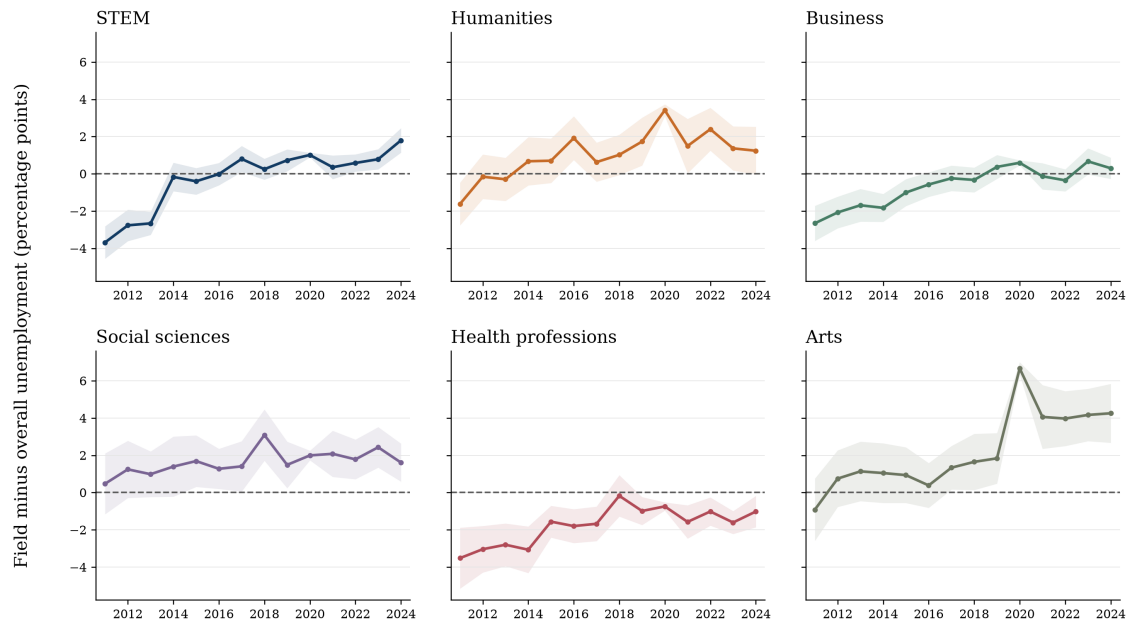
Notes: ACS one-year national person PUMS, 2011–2024; civilian, noninstitutional bachelor's-or-higher degree holders aged 18–25. Shares use the first-listed bachelor's field and divide by all selected degree holders. Shading shows 95% confidence intervals using the 80 person replicate weights. 2020 uses Census experimental weights. Field definitions and sample construction are described above.

Table A6: Changes in Bachelor's-Field Shares among Young Degree Holders

Field	2011 (%)	2024 (%)	Change (pp)
STEM	19.75	27.91	8.15
Humanities	9.50	5.50	–4.00
Business	19.15	17.66	–1.50
Social sciences	8.24	6.77	–1.47
Health professions	6.78	9.20	2.42
Other fields	36.57	32.97	–3.60

Notes: Sample, weights, and field definitions match Figure A9. Changes use unrounded shares; “Other fields” is the remainder and is not displayed in the figure.

Figure A10: Unemployment among Young College Graduates by Bachelor's Field

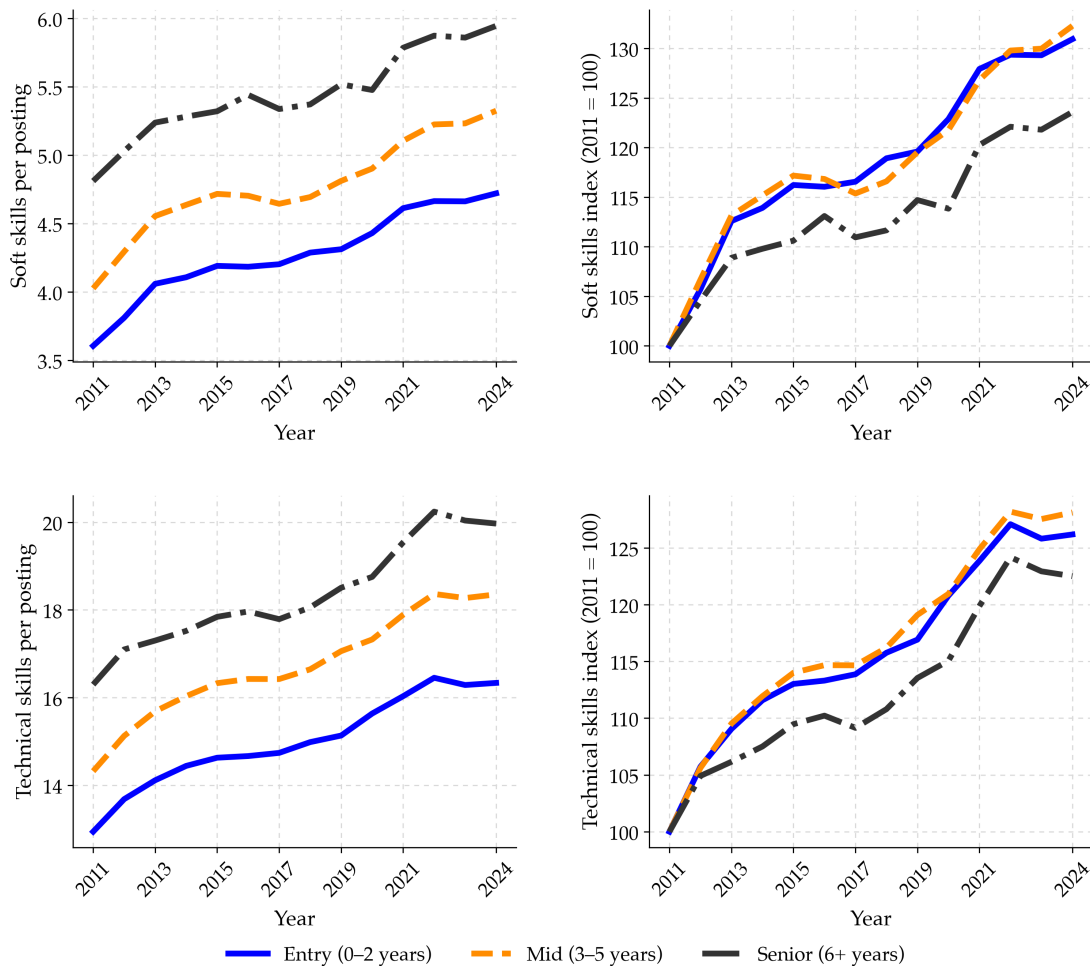


Notes: ACS one-year national person PUMS, 2011–2024. Lines show each field's unemployment rate minus the overall rate, in percentage points. The dashed zero line denotes equality with the overall rate. Field samples are civilian, noninstitutional bachelor's-or-higher degree holders aged 18–25, classified by their first-listed bachelor's field. Definitions match Figure A9, with arts (Census prefix 60) added. The overall benchmark includes civilian, noninstitutional respondents aged 16–55 across all education levels. Rates divide weighted unemployed respondents by weighted employed plus unemployed respondents; those outside the labor force are excluded from this denominator. Employment includes self-employment. Shading shows 95% confidence intervals for the gaps using the 80 person replicate weights, retaining covariance with the overall benchmark. 2020 uses Census experimental weights. Annual series are not smoothed.

A4.2 Skill Requirements by Required Work Experience

Figure A11 compares the level and growth of skill requirements across entry, mid-level, and senior postings, holding occupational composition fixed. The groups refer to the minimum experience stated in an advertisement, rather than the age or experience of the worker ultimately hired.

Figure A11: Skill Requirements by Required Work Experience



Notes: The sample consists of Lightcast postings requiring a bachelor's degree or higher and classified as full-time (more than 32 hours), 2011–2024. Entry, mid-level, and senior postings require 0–2, 3–5, and at least 6 years of experience, respectively. Postings with missing experience are excluded. The left panels show average skill counts per posting; the right panels index each series to 100 in 2011, rather than reporting annual growth rates. Soft skills appear in the top row and technical skills in the bottom row. Skill and posting counts are summed within occupation, experience group, and year before calculating ratios. All series use the same 473 occupations, observed in every group and year, and fixed occupation weights based on pooled posting shares across groups and years. The common sample contains 55,651,268 postings. The series are not adjusted for posting length and do not measure workers' skill acquisition.

A4.3 Unemployment Rate of Non-College Graduates

In Figure 2 in Section II, we showed that the unemployment rate of college graduates has increased in recent times. On the other hand, Figure A12 plots the unemployment rate of young

non-college workers against the overall unemployment rate. We observe that the unemployment rate of this group since 2016 is lower than at any other time since 1981. [Figure A12](#) also shows that the gap between the two lines is smaller during the same time period. One reason why non college graduates do not experience the same rise in their unemployment rate that college graduates do, is because they have been in the labor market for a longer time. This means that they may have already accumulated the level of work experience and soft skills by the age of 25 that employers ask for.

Figure A12: Young Non-College Workers vs. Overall Unemployment Rate

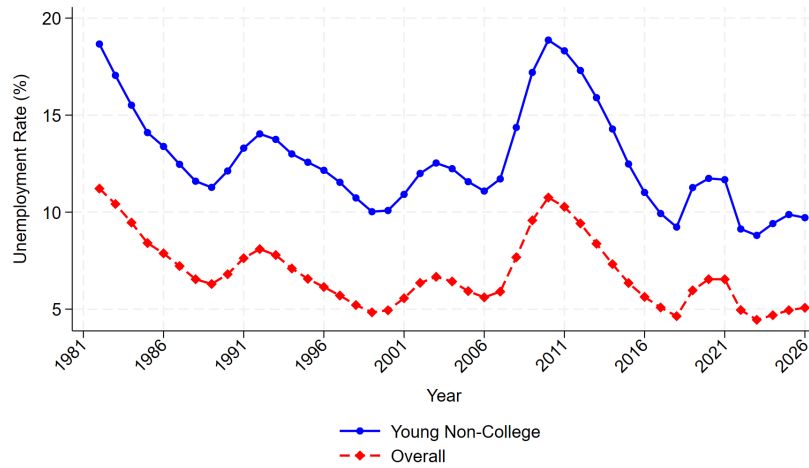
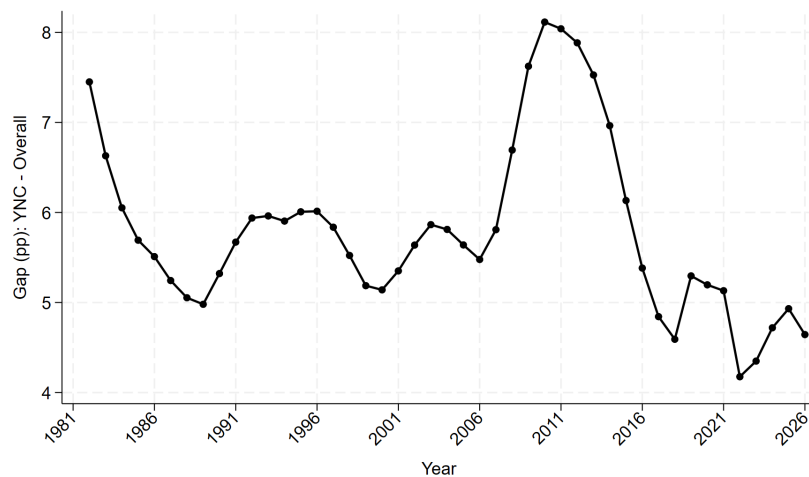


Figure A13: Gap Between The Young Non-College Workers vs. Overall Unemployment Rate



A4.4 Returns To Soft And Technical Skills For Young Workers

The soft-skill interaction is negative in each specification in [Table A7](#). For college workers it is about -3 log points per additional soft skill across specifications; the age-control estimate has $p = 0.064$. The college technical-skill interaction is essentially zero when occupation fixed

effects are included. The two skill dimensions therefore do not support equally stable conclusions across specifications. These estimates describe wage associations among employed workers and do not identify how workers acquire skills or the effect of skill requirements on unemployment.

Table A7: Young–Older Wage Differences and Skill Requirements: Robustness

	College		Noncollege	
	Young × Soft	Young × Tech.	Young × Soft	Young × Tech.
Baseline	-3.00** (1.42)	0.93*** (0.33)	-2.52*** (0.95)	-0.80*** (0.27)
Fixed occupation requirements	-3.29** (1.58)	0.78** (0.32)	-3.45*** (1.10)	-0.92*** (0.30)
Occupation fixed effects	-3.25*** (0.69)	0.00 (0.18)	-1.73*** (0.66)	-1.40*** (0.29)
Age and age squared	-2.61* (1.41)	0.85*** (0.33)	-2.22*** (0.81)	-1.10*** (0.26)
Observations	572,091		931,070	

Notes: The outcome is log real hourly earnings. Coefficients and standard errors are multiplied by 100; a coefficient is the change in the young–older log wage gap, in log points, associated with one additional advertised skill per posting. The saved sample covers 2011–2024 and ages 16–54; young denotes age 25 or younger and college denotes a bachelor’s degree or higher. These are the existing wage specifications, not the separate ages 21–29 employment diagnostics. All regressions include both skill measures, their interactions with young, sex, race, education, and calendar-month fixed effects. They use composition-adjusted final CPS weights and occupation-clustered standard errors. Each alternative changes only the stated feature of the baseline. Fixed occupation requirements pool men- tions and postings over 2011–2024. Occupation fixed effects absorb occupation mean wages; they do not absorb occupation-specific age wage gaps. The age specification adds age and age squared. Allocated earnings follow the existing wage extract. The coefficients, standard errors, and p-values come from the saved regression export; no model has been re-estimated for this table. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

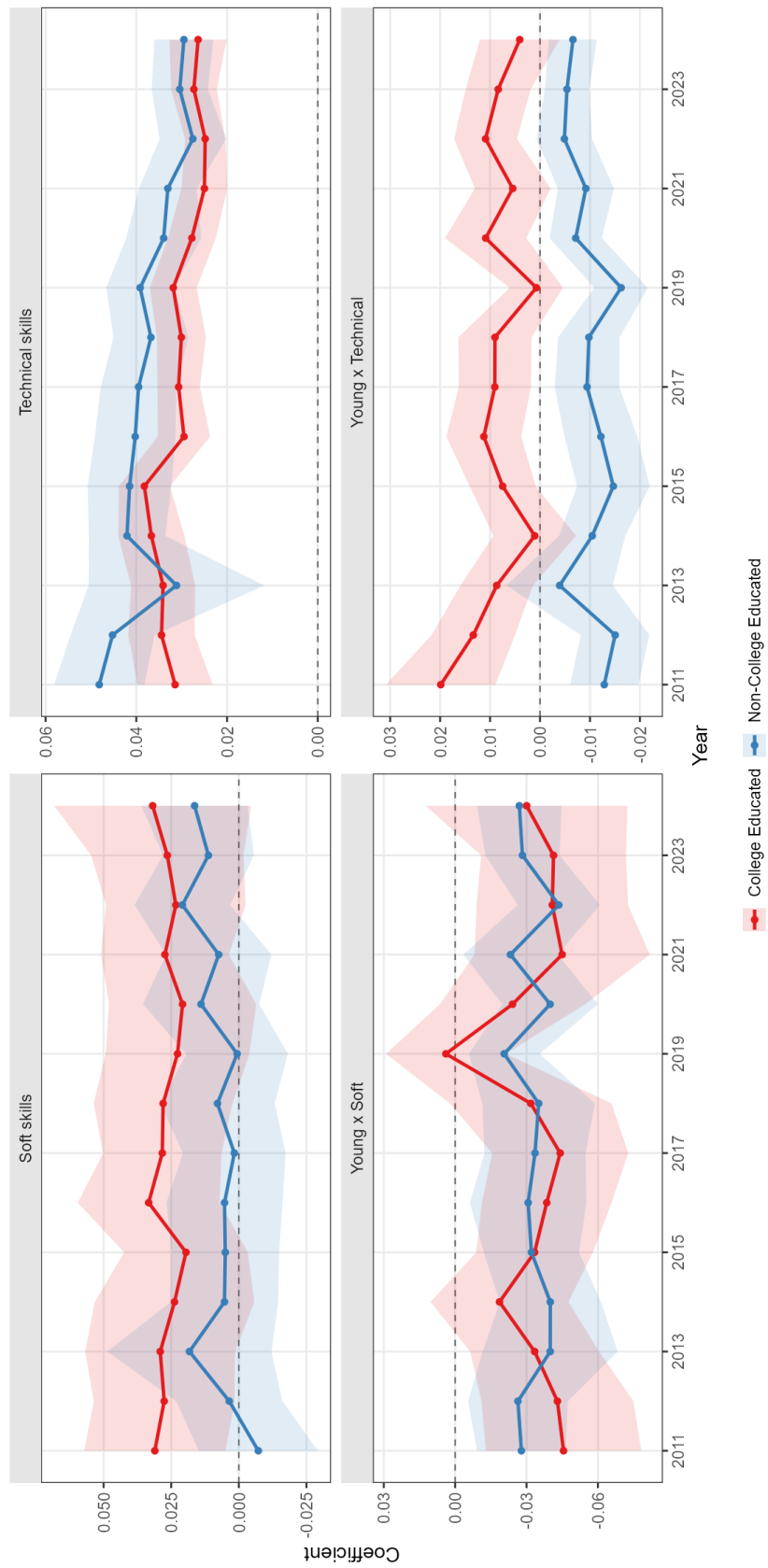
A5 Labor Market Flows of Young College Graduates

A deterioration in young graduates’ relative labor market outcomes can arise through slower job finding, more frequent separations, or changes in labor force participation. Using matched monthly CPS observations, we examine these margins over 2011–2024. We compare college graduates aged 21–29 with graduates aged 30–55 and workers aged 21–29 without a bachelor’s degree. Age and education are measured before the transition. Employment includes both wage work and self-employment. Each transition probability conditions on the worker’s initial labor market state.¹⁸

Young college graduates lose their initial advantage in finding employment from unem- ployment. Figure A15 shows the annual series, and Table A8 reports period averages. In 2011–2015, 28% of unemployed young graduates were employed the following month, com- pared with 21% of older graduates and 20% of young workers without a college degree. By

¹⁸We link respondents using the validated CPS person identifier, requiring consecutive calendar months and the appropriate progression through rotation groups. We exclude ASEC records and observations with missing labor market states or education, invalid demographic matches, or nonpositive longitudinal weights. Monthly rates use the origin month’s forward longitudinal weight, LNKFW1MWT. Annual and period estimates give each month equal weight. The age range differs from the main text’s definition of young workers as those aged 25 or younger.

Figure A14: Young Non-College Workers vs. Overall Unemployment Rate



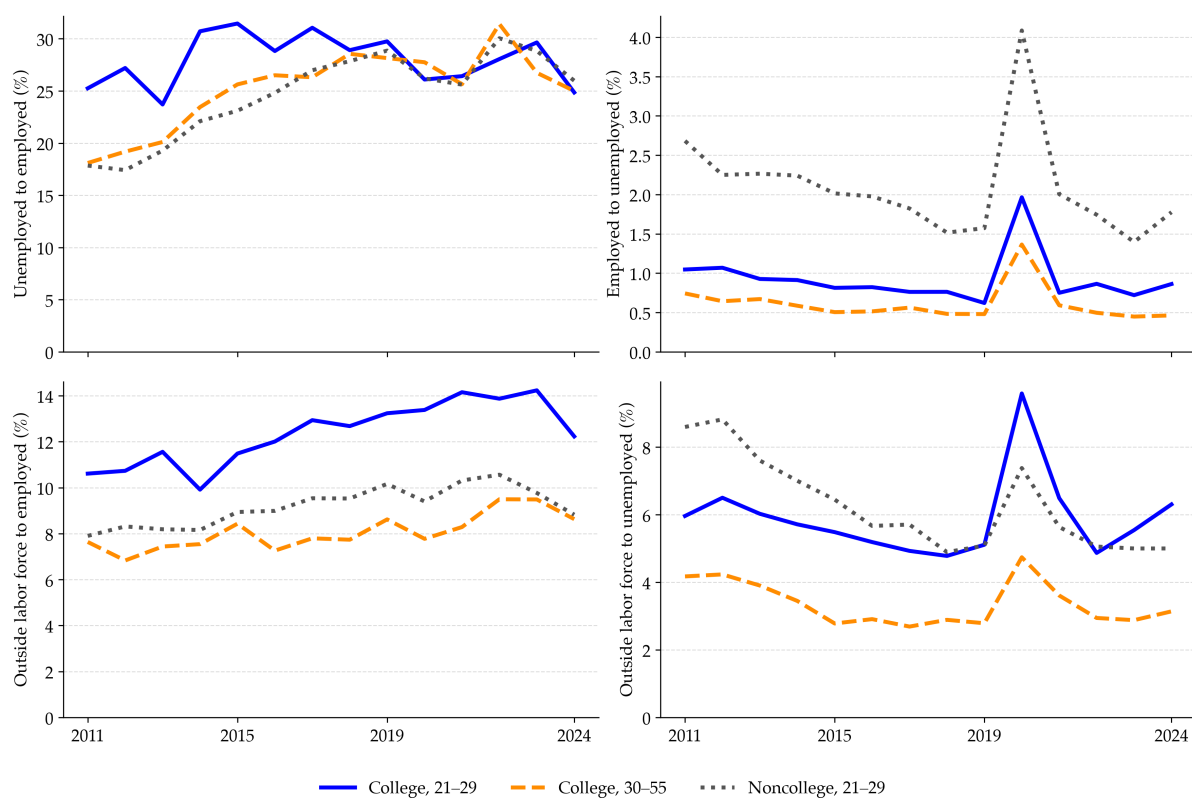
2022–2024, all three rates were about 28%. Young graduates’ job-finding rate was nearly unchanged across these periods, while both comparison groups improved. Their advantage over older graduates had already narrowed from 6 to 2 percentage points by 2016–2019, before the pandemic.

Separations into unemployment show little change in young graduates’ position relative to older graduates. The monthly employment-to-unemployment probability declined for both young and older graduates, remaining less than 1% for both groups in each period. The gap between them remained close to one-third of a percentage point. Young graduates also continued to have lower separation rates than young workers without a degree, although this difference narrowed. These transitions include both voluntary and involuntary separations.

The participation flows place a further limit on the interpretation. Among young graduates outside the labor force, the probability of moving directly into employment rose from 11% to 13%. These moves include both first entry and reentry, so they do not measure school-to-work transitions alone. The erosion of young graduates’ relative job finding is therefore specific to those starting from unemployment; it does not extend to all routes into employment.

The decline in young graduates’ job-finding advantage motivates examining why they have not shared in the improvement experienced by other unemployed workers. Changes in the composition of unemployment and differences in responses to the business cycle can also contribute to this pattern. The comparisons are descriptive, without confidence intervals, and do not identify the contribution of changing skill requirements to relative unemployment.

Figure A15: Labor Market Transitions by Age and Education



Notes: Annual averages of monthly transition probabilities in the matched CPS, 2011–2024. Each rate is the longitudinally weighted share of respondents in the initial state who enter the destination state the following month. College denotes a bachelor's degree or higher; employment includes self-employment.

Table A8: Monthly Labor Market Transition Probabilities

	2011–2015	2016–2019	2020–2021	2022–2024
<i>Unemployed → employed</i>				
College, 21–29	27.67	29.63	26.25	27.52
College, 30–55	21.29	27.38	26.70	27.72
Noncollege, 21–29	19.95	27.13	25.90	28.29
<i>Employed → unemployed</i>				
College, 21–29	0.95	0.74	1.36	0.82
College, 30–55	0.63	0.51	0.98	0.47
Noncollege, 21–29	2.29	1.72	3.04	1.64
<i>Outside labor force → employed</i>				
College, 21–29	10.86	12.72	13.77	13.45
College, 30–55	7.58	7.85	8.03	9.20
Noncollege, 21–29	8.30	9.55	9.86	9.72
<i>Outside labor force → unemployed</i>				
College, 21–29	5.94	5.00	8.04	5.57
College, 30–55	3.71	2.82	4.17	2.99
Noncollege, 21–29	7.69	5.34	6.51	5.02
<i>Employed → outside labor force</i>				
College, 21–29	1.94	1.93	2.01	1.97
College, 30–55	1.09	1.13	1.24	1.18
Noncollege, 21–29	3.46	3.63	4.24	3.74
<i>Unemployed → outside labor force</i>				
College, 21–29	16.76	18.95	18.65	18.07
College, 30–55	15.07	17.43	15.76	17.43
Noncollege, 21–29	23.85	25.71	24.70	25.04

Notes: Entries are percentages, averaged across months within each period. The matched CPS sample and weighting follow Figure A15. Each probability conditions on the initial state indicated by the row heading. The table reports all six transitions among employment, unemployment, and nonparticipation; pandemic years are shown separately.

B Skill Accumulation

We now extend the baseline model of [Section III](#) to let a worker's skill evolve over her career. While employed, skill rises by one rung each period with probability π ; while unemployed, it falls by one rung each period with probability q .¹⁹ Only the objects this affects are modified below; sections with no change point back to their counterpart in [Section III](#).

B1 Workers, Firms, and States

Firms are characterized by productivity $y \in \mathcal{Y} = [0, 1]$, fixed over time, and the aggregate state z_t evolves according to the Markov kernel $\Pi_z(z | z)$, exactly as in [Section III.A](#). The departure is that a worker's skill x is no longer fixed over her lifetime.

Skill lives on a linearly spaced grid $\mathcal{X} = \{\underline{x}, \underline{x} + \Delta, \dots, \bar{x}\}$. While employed, in each period a worker climbs one grid point with probability π and remains at her current rung with probability $1 - \pi$; the top rung $\bar{x}$ is absorbing. While unemployed, a worker falls one grid point with probability q and remains with probability $1 - q$; the bottom rung $\underline{x}$ is reflecting.

Workers exit the labor force at rate ξ and are replaced one for one by entrants, who enter unemployed at the lowest rung:

$$n_t(x) = \xi \mathbf{1}\{x = \underline{x}\}. \quad (30)$$

Because there is now entry and exit at every rung, the worker-skill distribution $\ell(x)$ is no longer conserved rung by rung as it is in [Section III.A](#); the stationary distribution of x is an outcome of entry, climbing, decay, and exit, rather than a primitive of the model.

B2 Production and Flow Value of Unemployment

The production function (1) and the flow value of unemployment (2) are unchanged in functional form. The only difference is that x is no longer a fixed lifetime trait: writing x_t for a given worker's rung at date t , her output $p_t(x_t, y, z)$ and flow value of unemployment $b_t(x_t)$ now move over her career as she climbs (while employed) or decays (while unemployed).

B3 Labor Markets

The stock identity $u_t(x) + \int h_t(x, y) dy = \ell(x)$ no longer holds rung by rung: mass now flows in at $\underline{x}$ (entry), out at every rung (exit), up a rung (climbing, employed only), and down a rung (decay, unemployed only). Total mass is still conserved, $\int [u_t(x) + \int h_t(x, y) dy] dx = 1$. The effective labor force (3), the matching function (4), and the job finding and filling rates (5)–(6) keep their exact functional form; only the distributions $u_t^+(x)$, $h_t^+(x, y)$ they are built from now reflect these additional flows ([Section B6](#)).

B4 Values and Surplus

For any function f of the rung, define the expectations over the worker's own climb and decay draws,

$$\mathbb{E}_\pi f(x) \equiv \pi f(x + \Delta) + (1 - \pi)f(x), \quad \mathbb{E}_q f(x) \equiv q f(x - \Delta) + (1 - q)f(x). \quad (31)$$

¹⁹We use π rather than p for the climbing probability to avoid clashing with the production function $p_t(x, y, z)$ of [Equation 1](#).

These are expectations over the worker's own idiosyncratic climb/decay draw, independent of the aggregate shock z_{t+1} that $\mathbb{E}_t$ averages over, so $\mathbb{E}_t$ commutes with both $\mathbb{E}_\pi$ and $\mathbb{E}_q$.

B4.1 Unemployment

Let $B_t(x)$ denote the value of unemployment for a worker at rung x . As in [Section III.D](#), if a worker remains unemployed throughout period t she receives $b_t(x)$, and because she has zero bargaining power, whether or not she receives a job offer in period $t + 1$ does not affect the value she holds: any firm matched with her offers exactly her reservation value at whatever rung she occupies. The departure from [Equation 8](#) is twofold: exit discounts the continuation at rate ξ , and her rung itself can fall while she is unemployed. With probability q she falls to $x - \Delta$ and with probability $1 - q$ she remains at x , so

$$B_t(x) = b_t(x) + \beta(1 - \xi) \left[q \mathbb{E}_t B_{t+1}(x - \Delta) + (1 - q) \mathbb{E}_t B_{t+1}(x) \right], \quad (32)$$

with no $x - \Delta$ term at the reflecting floor, $B_t(\underline{x}) = b_t(\underline{x}) + \beta(1 - \xi) \mathbb{E}_t B_{t+1}(\underline{x})$. Equivalently, using $\mathbb{E}_q$ from [\(31\)](#),

$$B_t(x) = b_t(x) + \beta(1 - \xi) \mathbb{E}_t \left[\mathbb{E}_q B_{t+1}(x) \right]. \quad (33)$$

When $q = 0$, [\(32\)](#) collapses to the closed form $B(x) = b(x)/(1 - \beta(1 - \xi))$, the direct analogue of [\(8\)](#) with the exit rate added to the discount. When $q > 0$ it is instead solved by a forward recursion from the floor up, since $B(x)$ depends on the already-solved $B(x - \Delta)$.

B4.2 Surplus

Let $P_t(x, y, z)$ denote the joint value of a match, with surplus defined as in [\(9\)](#), $S_t(x, y, z) = P_t(x, y, z) - B_t(x)$. Skill accumulation happens after current production and before next period's shocks are realized, so by the time the separation and poaching stages of [\(11\)](#) operate, the worker's rung is already $x + \Delta$ (with probability π) or x (with probability $1 - \pi$). The climb mixture therefore has to wrap the entire expression, not just one term of it:

$$\begin{aligned} P_t(x, y, z) = & p_t(x, y, z) \\ & + \beta(1 - \xi) \mathbb{E}_t \left[\pi \cdot \left(\left(1 - (1 - \delta) \mathbf{1}\{S_{t+1}(x + \Delta, y, z) \geq 0\} \right) B_{t+1}(x + \Delta) \right. \right. \\ & \quad \left. \left. + (1 - \delta) \mathbf{1}\{S_{t+1}(x + \Delta, y, z) \geq 0\} P_{t+1}(x + \Delta, y, z) \right) \right. \\ & \quad \left. + (1 - \pi) \cdot \left(\left(1 - (1 - \delta) \mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} \right) B_{t+1}(x) \right. \right. \\ & \quad \left. \left. + (1 - \delta) \mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} P_{t+1}(x, y, z) \right) \right]. \end{aligned} \quad (34)$$

Equivalently, using $\mathbb{E}_\pi$ from [\(31\)](#) to wrap the entire bracket as a single operator,

$$\begin{aligned} P_t(x, y, z) = & p_t(x, y, z) \\ & + \beta(1 - \xi) \mathbb{E}_t \mathbb{E}_\pi \left[\left(1 - (1 - \delta) \mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} \right) B_{t+1}(x) \right. \\ & \quad \left. + (1 - \delta) \mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} P_{t+1}(x, y, z) \right]. \end{aligned} \quad (35)$$

Within each branch the worker's rung is fixed, so the Bertrand argument of [Section III.D](#) applies unchanged: whichever firm she ends up at, she realizes exactly $P_{t+1}(\cdot, y, z)$, so the competing-offer integral collapses the same way it does in (11). Combining (32), (9), and (34), and writing the two Bellman equations side by side rather than netting P against the flow $b_t(x)$, gives the match surplus

$$\begin{aligned} S_t(x, y, z) = & p_t(x, y, z) - b_t(x) \\ & + \beta(1 - \xi) \left[\pi \mathbb{E}_t B_{t+1}(x + \Delta) + (1 - \pi) \mathbb{E}_t B_{t+1}(x) - q \mathbb{E}_t B_{t+1}(x - \Delta) - (1 - q) \mathbb{E}_t B_{t+1}(x) \right] \\ & + \beta(1 - \xi)(1 - \delta) \mathbb{E}_t \left[\pi \mathbf{1}\{S_{t+1}(x + \Delta, y, z) \geq 0\} (P_{t+1}(x + \Delta, y, z) - B_{t+1}(x + \Delta)) \right. \\ & \quad \left. + (1 - \pi) \mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} (P_{t+1}(x, y, z) - B_{t+1}(x)) \right], \end{aligned} \quad (36)$$

where the continuation term is written in terms of P_{t+1} and B_{t+1} directly rather than S_{t+1} , using $\mathbf{1}\{S \geq 0\} \cdot S \equiv \max\{0, S\}$ together with $S_{t+1} = P_{t+1} - B_{t+1}$ – the viability condition itself is still naturally stated in terms of S . Writing both brackets using $\mathbb{E}_\pi$ and $\mathbb{E}_q$ from (31), the surplus can be written compactly as

$$\begin{aligned} S_t(x, y, z) = & p_t(x, y, z) - b_t(x) \\ & + \beta(1 - \xi) \mathbb{E}_t \left[\mathbb{E}_\pi \left((1 - \delta) \mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} (P_{t+1}(x, y, z) - B_{t+1}(x)) \right) \right] \\ & + \beta(1 - \xi) \mathbb{E}_t \left[\mathbb{E}_\pi B_{t+1}(x) - \mathbb{E}_q B_{t+1}(x) \right]. \end{aligned} \quad (37)$$

Relative to (12), three things change: (i) the discount $\beta(1 - \delta)$ becomes $\beta(1 - \xi)(1 - \delta)$, since exit now discounts the match as well; (ii) the continuation becomes a mixture over the climbed and un-climbed rung rather than a single term; and (iii) a genuinely new term, $\beta(1 - \xi) \mathbb{E}_t [\mathbb{E}_\pi B_{t+1}(x) - \mathbb{E}_q B_{t+1}(x)]$ – equivalently $\beta(1 - \xi) (\pi \mathbb{E}_t [B_{t+1}(x + \Delta) - B_{t+1}(x)] + q \mathbb{E}_t [B_{t+1}(x) - B_{t+1}(x - \Delta)])$ – appears, which has no counterpart when x is fixed for life: one piece is the option value of climbing while employed, the other is the option value of avoiding decay by staying employed rather than becoming unemployed. This last term accounts for the large majority of the effect of skill accumulation on the surplus: the channel one might expect, that the match becomes more productive next period because the worker may be more skilled, is a minor part of the total effect by comparison. Proposition 1 of [Lise and Robin \(2017\)](#), that the match surplus depends on time only through the aggregate productivity shock, continues to hold here.

B5 Vacancy Posting

The vacancy cost (13), the value of a filled vacancy (14), the vacancy-posting condition (15), and equilibrium tightness (17) are unchanged in form; $J_t(y)$ is still built by integrating $\max\{S_t(x, y, z), 0\}$ against the current $u_t^+(x)$, $h_t^+(x, y)$, which now carry the entry, exit, climbing, and decay flows of [Section B6](#).

B6 Worker Flows

Each period now has three stages. Separations and matching operate first, exactly as in [Section III.F](#), since they depend on the surplus $S_t(x, y, z)$ evaluated at the rung x a worker holds at

the start of the period; entry, exit, climbing, and decay are applied together last, after matching, since they move a worker to a new rung only once the period's matches have been determined.

Separations operate exactly as in (18)–(19):

$$u_t^+(x) = u_t(x) + \int_y \left(\delta + (1 - \delta) \mathbf{1}\{S_t(x, y, z) < 0\} \right) h_t(x, y) dy, \quad (38)$$

$$h_t^+(x, y) = (1 - \delta) \mathbf{1}\{S_t(x, y, z) \geq 0\} h_t(x, y). \quad (39)$$

Matching and poaching then operate exactly as in (20)–(21). Because the skill transition has not yet occurred, the resulting distributions are not yet u_{t+1} , h_{t+1} ; call them $u_{t+1/2}(x)$ and $h_{t+1/2}(x, y)$:

$$u_{t+1/2}(x) = u_t^+(x) - u_t^+(x) \lambda_{w,t} \int \frac{v_t(y)}{V_t} \mathbf{1}\{S_t(x, y, z) \geq 0\} dy, \quad (40)$$

$$\begin{aligned} h_{t+1/2}(x, y) = & h_t^+(x, y) \left[1 - \int s \lambda_{w,t} \frac{v_t(y')}{V_t} \mathbf{1}\{S_t(x, y, z) < S_t(x, y', z)\} dy' \right] \\ & + \int h_t^+(x, y') s \lambda_{w,t} \frac{v_t(y)}{V_t} \mathbf{1}\{S_t(x, y, z) > S_t(x, y', z)\} dy' \\ & + u_t^+(x) \lambda_{w,t} \frac{v_t(y)}{V_t} \mathbf{1}\{S_t(x, y, z) \geq 0\}. \end{aligned} \quad (41)$$

Entry, exit, and the skill transition are applied last, combined into a single step: exit removes a fraction ξ of every rung, and entrants replace that mass one for one at the bottom rung $\underline{x}$, consistent with (30). An employed worker who does not exit climbs one rung with probability π ; an unemployed worker who does not exit falls one rung with probability q . Writing the interior case first, then the lowest rung, then the highest,

$$h_{t+1}(x, y) = \begin{cases} (1 - \xi) \left[\pi h_{t+1/2}(x - \Delta, y) + (1 - \pi) h_{t+1/2}(x, y) \right], & \underline{x} < x < \bar{x}, \\ (1 - \xi)(1 - \pi) h_{t+1/2}(\underline{x}, y), & x = \underline{x}, \\ (1 - \xi) \left[h_{t+1/2}(\bar{x}, y) + \pi h_{t+1/2}(\bar{x} - \Delta, y) \right], & x = \bar{x}, \end{cases} \quad (42)$$

where the top case reflects that the top rung is absorbing: all the mass already there stays (not just the $1 - \pi$ fraction that fails to climb), and it also receives the mass that would otherwise have climbed past it. The unemployed distribution is the mirror image, with decay in place of climbing and a reflecting floor in place of the absorbing ceiling, plus the entrant inflow at $\underline{x}$:

$$u_{t+1}(x) = \begin{cases} (1 - \xi) \left[q u_{t+1/2}(x + \Delta) + (1 - q) u_{t+1/2}(x) \right], & \underline{x} < x < \bar{x}, \\ (1 - \xi) \left[u_{t+1/2}(\underline{x}) + q u_{t+1/2}(\underline{x} + \Delta) \right] + \xi, & x = \underline{x}, \\ (1 - \xi)(1 - q) u_{t+1/2}(\bar{x}), & x = \bar{x}. \end{cases} \quad (43)$$

Both operators conserve mass: (40)–(41) leave total mass at 1, applying the factor $1 - \xi$ to every rung removes a total mass of ξ , and adding ξ back at $\underline{x}$ in (43) restores it.

B7 Aggregate Quantities

Aggregate output (22) and aggregate consumption (23) are unchanged in form; both integrate against the h_{t+1}, u_{t+1} of Section B6 rather than those of Section III.F.

B8 Equilibrium

The equilibrium of Section III.H is unchanged in its six defining conditions. Condition 1 now refers to the Bellman equations (32) and (36), and condition 5 now includes the separations, matching, entry, exit, climbing, and decay operators of Section B6 as part of the law of motion for u_t and h_t .

B9 Wages

The share of surplus $\sigma_t(x, y, z)$ is defined exactly as in (25). As in the base model, σ_t is renegotiated only when a competing offer arrives; climbing a rung is not itself a renegotiation event, so if no outside offer arrives next period the worker keeps the same share σ_t at her new rung. The worker's value from employment $W_t(x, y, z, \sigma)$ extends (26) the same way P_t extends (11) in (34): the climb mixture wraps the entire continuation, evaluated at the climbed or un-climbed rung, since by the start of $t+1$ the worker's rung is already $x+\Delta$ (probability π) or x (probability $1-\pi$):

$$\begin{aligned}
 W_t(x, y, z, \sigma) = & w_t(x, y, z, \sigma) \\
 & + \beta(1-\xi) \mathbb{E}_t \left[\pi \left(\delta + (1-\delta) \mathbf{1}\{S_{t+1}(x+\Delta, y, z) < 0\} \right) B_{t+1}(x+\Delta) \right. \\
 & \quad \left. + (1-\pi) \left(\delta + (1-\delta) \mathbf{1}\{S_{t+1}(x, y, z) < 0\} \right) B_{t+1}(x) \right] \\
 & + \beta(1-\xi)(1-\delta) \mathbb{E}_t \left[\pi \mathbf{1}\{S_{t+1}(x+\Delta, y, z) \geq 0\} \left(1 - s\lambda_{w,t+1}(y) \right) W_{t+1}(x+\Delta, y, z, \sigma) \right. \\
 & \quad \left. + (1-\pi) \mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} \left(1 - s\lambda_{w,t+1}(y) \right) W_{t+1}(x, y, z, \sigma) \right] \\
 & + \beta(1-\xi)(1-\delta) \mathbb{E}_t \left[\pi \mathbf{1}\{S_{t+1}(x+\Delta, y, z) \geq 0\} s\lambda_{w,t+1}(y) \right. \\
 & \quad \times \int \min \left\{ P_{t+1}(x+\Delta, y, z), \max \{ W_{t+1}(x+\Delta, y, z, \sigma), P_{t+1}(x+\Delta, y', z) \} \right\} \frac{v_{t+1}(y')}{V_{t+1}} dy' \\
 & \quad + (1-\pi) \mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} s\lambda_{w,t+1}(y) \\
 & \quad \times \int \min \left\{ P_{t+1}(x, y, z), \max \{ W_{t+1}(x, y, z, \sigma), P_{t+1}(x, y', z) \} \right\} \frac{v_{t+1}(y')}{V_{t+1}} dy' \Big].
 \end{aligned} \tag{44}$$

Equivalently, using $\mathbb{E}_\pi$ from (31) to wrap each bracket as a single operator,

$$\begin{aligned}
W_t(x, y, z, \sigma) = & w_t(x, y, z, \sigma) \\
& + \beta(1 - \xi) \mathbb{E}_t \mathbb{E}_\pi \left[\left(\delta + (1 - \delta) \mathbf{1}\{S_{t+1}(x, y, z) < 0\} \right) B_{t+1}(x) \right] \\
& + \beta(1 - \xi)(1 - \delta) \mathbb{E}_t \mathbb{E}_\pi \left[\mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} (1 - s \lambda_{w,t+1}(y)) W_{t+1}(x, y, z, \sigma) \right] \\
& + \beta(1 - \xi)(1 - \delta) \mathbb{E}_t \mathbb{E}_\pi \left[\mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} s \lambda_{w,t+1}(y) \right. \\
& \quad \times \int \min \{ P_{t+1}(x, y, z), \max \{ W_{t+1}(x, y, z, \sigma), P_{t+1}(x, y', z) \} \} \frac{v_{t+1}(y')}{V_{t+1}} dy' \Big].
\end{aligned} \tag{45}$$

As in (26), the four terms are: current wages; the value if the match is destroyed next period (at whichever rung she climbed to); the value of staying with no outside offer; and the value when an outside offer arrives, split the same way as (34) into the climbed-rung and un-climbed-rung branches. As with S_t , q does not appear directly here; it only reaches W_t through B_{t+1} .

Rearranging (45) using (25) to write W_{t+1} in terms of P_{t+1} and B_{t+1} , and substituting (34) and (32), gives the wage equation by exactly the same steps as the base model's derivation of (27) – applied inside the $\mathbb{E}_\pi$ wrapper, since every continuation in (45) carries the same climb mixture. The one place the derivation departs from the base model is that (32) pins down the $\mathbb{E}_q$ -weighted combination of B_{t+1} , not the $\mathbb{E}_\pi$ -weighted combination that appears here; writing $\mathbb{E}_\pi B_{t+1}(x) = \mathbb{E}_q B_{t+1}(x) + (\mathbb{E}_\pi B_{t+1}(x) - \mathbb{E}_q B_{t+1}(x))$ to use (32) on the first piece leaves the second piece as a residual that survives into the final wage equation:

$$\begin{aligned}
w_t(x, y, z, \sigma) = & \sigma_t p_t(x, y, z) + (1 - \sigma_t) b_t(x) \\
& - \beta(1 - \xi)(1 - \sigma_t) \mathbb{E}_t \left[\mathbb{E}_\pi B_{t+1}(x) - \mathbb{E}_q B_{t+1}(x) \right] \\
& - \beta(1 - \xi)(1 - \delta) \mathbb{E}_t \mathbb{E}_\pi \left[s \lambda_{w,t+1} \mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} \Omega_{t+1}(x, y, z, \sigma) \right],
\end{aligned} \tag{46}$$

where the renegotiation term $\Omega_t(x, y, z, \sigma)$ is unchanged in form from (28),

$$\Omega_t(x, y, z, \sigma) = \int \left[\min \left\{ S_t(x, y, z), \max \left\{ S_t(x, y', z), \sigma_t S_t(x, y, z) \right\} \right\} - \sigma_t S_t(x, y, z) \right] \frac{v(y')}{V} dy', \tag{47}$$

evaluated at whichever rung – x or $x + \Delta$ – the surrounding $\mathbb{E}_\pi$ in (46) is currently averaging over. As in (28), writing $S \equiv S_t(x, y, z)$ and $S' \equiv S_t(x, y', z)$, the bracketed integrand reduces to

$$\min \{ S, \max \{ S', \sigma_t S \} \} - \sigma_t S = \begin{cases} 0, & S' \leq \sigma_t S \quad (\text{offer below current wage}) \\ S' - \sigma_t S, & \sigma_t S < S' \leq S \quad (\text{worker stays, renegotiates up}) \\ (1 - \sigma_t) S, & S' > S \quad (\text{worker leaves, firm loses its full share}) \end{cases}$$

Equivalently, $\Omega_t(x, y, z, \sigma)$ can be written in terms of P_t and B_t instead of S_t . Substituting $S_t(x, y, z) = P_t(x, y, z) - B_t(x)$ and $S_t(x, y', z) = P_t(x, y', z) - B_t(x)$ into (47) and factoring $B_t(x)$ out of the inner max,

$$\max \{ S_t(x, y', z), \sigma_t S_t(x, y, z) \} = \max \left\{ P_t(x, y', z), \sigma_t P_t(x, y, z) + (1 - \sigma_t) B_t(x) \right\} - B_t(x),$$

and using (25) to recognize $\sigma_t P_t(x, y, z) + (1 - \sigma_t) B_t(x) = B_t(x) + \sigma_t S_t(x, y, z) = W_t(x, y, z, \sigma)$, the $B_t(x)$ terms collapse (exactly as in the derivation of (46) itself) to leave

$$\Omega_t(x, y, z, \sigma) = \int \left[\min \{ P_t(x, y, z), \max \{ P_t(x, y', z), W_t(x, y, z, \sigma) \} \} - W_t(x, y, z, \sigma) \right] \frac{v(y')}{V} dy'. \quad (48)$$

As in (47), writing $P \equiv P_t(x, y, z)$, $P' \equiv P_t(x, y', z)$, and $W \equiv W_t(x, y, z, \sigma)$, the bracketed integrand of (48) reduces to

$$\min \{ P, \max \{ P', W \} \} - W = \begin{cases} 0, & P' \leq W \quad (\text{offer below current wage}) \\ P' - W, & W < P' \leq P \quad (\text{worker stays, renegotiates up}) \\ P - W, & P' > P \quad (\text{worker leaves, firm loses its full share}) \end{cases}$$

the same three cases as (47), since $P \geq W$ whenever the match is viable ($\sigma_t \leq 1$, $S_t(x, y, z) \geq 0$), and since $P - W = (1 - \sigma_t)S$ term by term with the S -based cases above.

A third, equivalent form nests the min and max the other way around, writing the ceiling $(1 - \sigma_t)S_t(x, y, z)$ as the outer bound and the renegotiation gain $P_t(x, y', z) - W_t(x, y, z, \sigma)$, floored at 0, as the inner term:

$$\Omega_t(x, y, z, \sigma) = \int \min \left\{ (1 - \sigma_t) S_t(x, y, z), \max \left\{ 0, P_t(x, y', z) - W_t(x, y, z, \sigma) \right\} \right\} \frac{v(y')}{V} dy'. \quad (49)$$

This reduces to the same three cases once more: if the competing offer is below $W_t(x, y, z, \sigma)$ the inner max is 0, so the worker gets 0; if it exceeds $W_t(x, y, z, \sigma)$ but the renegotiation gain is still below the ceiling $(1 - \sigma_t)S_t(x, y, z) = P_t(x, y, z) - W_t(x, y, z, \sigma)$, the worker gets the renegotiation gain itself; and once the competing offer pushes the gain past that ceiling, the worker is capped at $(1 - \sigma_t)S_t(x, y, z)$ – exactly the point at which she switches firms. Writing out $\mathbb{E}_\pi$ and $\mathbb{E}_q$ from (31), the new term is

$$-\beta(1 - \xi)(1 - \sigma_t) \left(\pi \mathbb{E}_t [B_{t+1}(x + \Delta) - B_{t+1}(x)] + q \mathbb{E}_t [B_{t+1}(x) - B_{t+1}(x - \Delta)] \right): \quad (50)$$

two separate pieces involving B_{t+1} , one weighted by the climbing probability π and one by the decay probability q , rather than a single π -weighted term.

Three things change relative to (27): the discount becomes $\beta(1 - \xi)(1 - \delta)$ wherever it multiplies a continuation conditional on the match surviving; the renegotiation term is evaluated under the same climb mixture $\mathbb{E}_\pi$ as the surplus; and a new term, (50), appears, unconditional on whether the match itself survives – the worker's own share, $(1 - \sigma_t)$, of exactly the negative of the option-value term in (37). Since B_{t+1} is increasing in the rung, $\mathbb{E}_\pi B_{t+1}(x) > \mathbb{E}_q B_{t+1}(x)$, so this term lowers the wage: the firm pays less today because the worker is already compensated, outside of wages, by the higher value of staying employed – climbing with probability π – relative to becoming unemployed and facing decay with probability q . q now enters (46) directly, through this term, not only through $b_t(x)$.